Contents

1 Open and closed sets

Definition. An interval on the number line is called open if it goes from A to B without containing the endpoints A and B, i.e. it does not contain its boundary.

Definition. $\mathbb{R}$ is the set of real numbers and $\mathbb{R}^n$ is the set corresponding to n-dimensional space

The same can be said for subsets of the plane or higher dimensional spaces – they are open if they do not contain any of their boundary. They are closed if they contain all of their boundary. The formal definition of a boundary is a point where no matter how small of a circle or ball you make around that point, it will contain points both inside and outside the set. Note that since an open set contains none of its boundary, it has the property that for any point in the open set, all points sufficiently close to that point are in that set, so open sets must always have some thickness to them.

This is related to the idea that there is no smallest real number greater than another real number, and whether sets are open or closed turns out to be surprisingly important. Also, arbitrary unions of open sets are open since each point has a "ball" around it inside one of the open sets in the union and thus the whole thing, similarly arbitrary intersections of closed sets are closed. The reverse implications only hold for finite unions and intersections (since infinitely many balls intersection could be a point but finitely many is a ball). Sets can be both open and closed (such as $\mathbb{R}$), and sets can be neither open nor closed.

A useful way to think of closed sets is as sets where the set of everything not in the closed set forms an open set.

The closure of a set is the union of a set with its boundary, or the intersection of all sets containing that set.

We usually call open intervals (a,b) meaning everything from a to b not inclusive, and closed intervals [a,b] to mean everything from a to b inclusive, and [a,b) means everything from a to b including a but not b.

Lemma. Suppose we have a sequence of closed and bounded sets $C_n \subseteq \mathbb{R}^k$ such that for all n we have $C_n \supseteq C_{n+1}$ and they are all non-empty, then their intersection is non-empty.

Proof. As we discuss later in this document, the Bolzano-Weierstrass theorem holds in $\mathbb{R}^k$ - bounded sequences have convergent subsequences. We will pick any sequence (a_n) such that $a_n \in C_n$, then this sequence is bounded as it is contained in C_1 so it has a convergent subsequence, and its limit is in C_j for every j since its elements are eventually all in C_j and C_j is closed.

□

Note that the statements "for any convergent sequence in the set its limit is in the set" and "The set is closed" are equivalent, as the limit of a sequence is either in the set or part of its boundary. Therefore if the set contains its entire boundary it contains limit points of every sequence, and if it is missing a boundary point then by definition of a boundary point there is a sequence in the set converging to that boundary point which does not have a limit point in the set.

2 Cardinality

Now, I will talk about cardinality. This is really fun stuff, but I will not go through all the fun results here, only the results we need. The motivation of this is to show that there are, in a precise sense, "more" real numbers than natural numbers, so I can show that you cannot take a sum over all the real numbers of positive numbers and still get a finite number.

We consider two sets to be the same size if there is a one to one correspondence between their elements. This sounds obvious, but it shows that there are equally as many even positive integers as positive integers, because we can map 1—2 and 2—4 and 3—6 and so on, giving us a one to one correspondence.

Theorem. We cannot find a one to one correspondence between real numbers on any non-degenerate interval and positive integers.

The proof of this theorem is very famous in "popular math" and the slogan is "there are different sizes of infinity".

Proof. We will do this for the interval (0,1) then note that we can get to any interval by scaling, or the entre reals by doing $-\cot(\pi x)$ for example. The property of there being a correspondence to positive integers is called being countable. But, I think listable is a better term.

So, suppose we do, in fact, have a list of all the real numbers between 0 and 1.

1. 0.**3**141592653589793...
2. 0.2**7**18281828459045...
3. 0.14**1**4213562373095...
4. 0.349**8**579345858968...
5. 0.9988**2**37478199283...

Now construct a new number where the n'th decimal digit after the point is not the red number, and such that we do not have infinite trailing 9's or 0's (to ensure the number has a unique decimal representation). For the numbers in our list we pick the one with trailing 0's and not trailing 9's whenever we have to make this choice). There are always many ways to do this. For example, adding (or subtracting if adding would get us a 9 or a 0) 1 to each bold digit we could have 0.48273... But this is clearly not on the list – It differs from everything on the list by at least one digit. Contradiction. Essentially, this makes it so if you think you found a way to list them, I can prove you are wrong.

□

We want to prove that you can't add an unlistable amount of real numbers greater than 0 and get a finite result.

Now if we had a sum over the real numbers of positive numbers x, split them into subsets:

$$x \geq 1, \frac{1}{2} \leq x < 1, \frac{1}{3} \leq x < \frac{1}{2}, \frac{1}{4} \leq x < \frac{1}{3}, ...$$

Then one of these subsets has to have infinitely many elements. Why? If they all had finitely many elements then they would form a subset of this infinite table:

	1	2	3	4	5	6	7	8	9	...
Set 1										
Set 2										
Set 3										
Set 4										
Set 5										
Set 6										
Set 7										
Set 8										
...										

So I could then go along the table in a zigzag pattern like this in the order shown below, adding to my list whenever the elements are in the table, and not adding them otherwise.

	1	2	3	4	5	6	7	8	9	...
Set 1	1	2	6	7	15	...				
Set 2	3	5	8	14						
Set 3	4	9	13							
Set 4	10	12								
Set 5	11									
Set 6										
Set 7										
Set 8										
...										

All cells get reached eventually, so this would imply we can list the cells. But we are assuming we are adding one term for each real number, and we cannot list the real numbers. Therefore, one of the sets

$$x \geq 1,\ \frac{1}{2} \leq x < 1,\ \frac{1}{3} \leq x < \frac{1}{2},\ \frac{1}{4} \leq x < \frac{1}{3}.$$

has infinitely many elements, so the sum is infinite.

Now here is the key point: The set of all integers is countable by the list $0, 1, -1, 2, -2, 3, -3, ...$ so the set of pairs of integers is countable. Each pair (except those that contain a 0 in the second position) corresponds to a rational number and all rational numbers are gotten this way, in fact infinitely many times. Therefore, the set of rational numbers is also countable. This is important because it means if we have an increasing function then it has at most countably many discontinuities, as each discontinuity goes "through" a rational number. We will use this fact a lot when proving the central limit theorem.

3 Measure theory

3.1 How my attempts to bypass this failed

You probably have noticed that this chapter is very long and intimidating. And to that, I completely agree, which is why I spent a year trying to prove the results we need without doing all of this. After a year, I'm convinced it's impossible, and I'm going to explain to you exactly how all my attempts failed because I know that by just showing you the end result, you are not going to understand this as well as I understand it now.

3.1.1 My first attempt

Definition. We say a measure on $\mathbb{R}$ is a way of assigning a length to subsets of $\mathbb{R}$, whatever that might look like. For example, the open interval $(0, 1)$ is something we would expect to have a length of 1.

Note that sets are said to be disjoint if any 2 have empty intersection. A measure is said to be countably additive is if for any sequence of disjoint sets (A_n), the measure of their union is the sum of their measure.

I did not explicitly show how to define a measure on $\mathbb{R}$. I cited intuition, thinking it was obvious like it's just the amount of mass. I later realized that this intuition was flawed, because as we will see, it is impossible to sensibly define a countably additive measure to all subsets of $\mathbb{R}$. I then proceeded to define simple functions - something that you need to have a working measure to even define - and defined the Lebesgue integral based on that. I then attempted to prove a core theorem, which is that if you have a sequence of increasing functions f_n with $f_n \to f$ everywhere, then $\int f_n \to \int f$. Although I had a convincing looking proof, the theorem as I stated it was false if you don't define length properly, since if your measure isn't countably additive and you want to have it so that the integral of a function that is 1 on that set and 0 everywhere else equals the measure of that set, you need your measure to be countably additive for that proposition to hold.

Note, also, that swapping limits and integrals like that is *not* obvious, for if $f_n(x)$ is defined as n for $0<x<1/n$ then the integral of this will be 1, but at all points between 0 and 1 f_n will eventually go to 0, so the integral of the limit is not the limit of the integral.

If there actually was a way to assign a countably additive measure to every set, I probably would have hand-waved it citing intuition since this is an A level document. It wouldn't be the only time I do that for true well known statements that are easier to get an intuition for than to prove. However, since that is false and we will disprove it in this chapter, my first attempt fell apart, since everything was built on the above theorem which was built on sand.

3.1.2 My second attempt

Definition. An indicator function of a set A is a function that is 1 on that set and 0 elsewhere. We often denote this 1_A.

When I started to learn measure theory I realized that my first attempt failed for the reason explained above. I was then made aware of a new interpretation of the integral. The interpretation was as follows. Essentially, you assign a measure to sets which are finite unions of intervals - as those have an obvious length. You then define a step function as finite weighted sums of indicator functions of such sets. You then say that any function that is the limit of an increasing sequence of step functions is integrable and the integral is the limit of the area of the step functions.

This turned out to work beautifully for proving the above theorem - that if you have a sequence of increasing functions f_n with $f_n \to f$ everywhere, then $\int f_n \to \int f$, as well as the dominated convergence theorem which is of a similar nature, and needed in the proof of the central limit theorem.

However, in order to make this approach actually work, I have to show that this integral is equal to the Riemann integral - this is the standard integral that we will define shortly. This was necessary to use facts such as the integral being the reverse of the derivative. After exhibiting a function that was Riemann integrable but not integrable in the above sense, I realized that this approach couldn't work. I wanted the standard integral to have some properties and I proved a different integral had those properties and couldn't prove they were the same. The only way to salvage the proof was to define step functions in terms of arbitrary measurable sets, which gets us straight back to the way my first attempt failed.

3.1.3 My third attempt

In my third attempt I tried to prove the properties I needed for the Riemann integral (see section 3.2 of this document) directly.

I was able to prove that for bounded continuous functions on bounded domains, integrals can be swapped (so $\iint f(x,y)\,dxdy = \iint f(x,y)\,dydx$) - there was no catch there. I was also able to prove that if $f_n \to f$ and f_n, f is Riemann integrable and all f_n have $|f_n| < g$ where the integral of the absolute value of g is finite then $\int f_n \to \int f$ for bounded functions on a finite bounded domain - again no catch there. However the catch is that this wasn't enough. I needed integration on unbounded domains and, more importantly, integration with respect to probability distributions. It is in the second point that a Riemann integral approach breaks down, since a naive integral does not work since a distribution need not have a density, and restricting to distributions with a density does not suffice as their limit need not have a density.

My attempt to fix the above issue involved introducing the Riemann-Stieltjes integral, where essentially you take a function g and replace $x_i - x_{i-1}$ in the definition below with $g(x_i) - g(x_{i-1})$. If g is a cdf, then this sort of does what we want, but we run into an issue where if g and the function we are integrating share a discontinuity point, the lower and upper integrals will never converge to the same value. If a cdf was discontinuous at all rational number points - which is possible - then this approach completely breaks down, as you cannot "just isolate" each discontinuity.

I also used the "fact" that if $|f|$ is Riemann integrable then so is f, which is false - for example f might be 1 at all rational numbers and -1 at all irrational numbers.

Overall, there were so many issues with this approach, that not only is this approach easier than fixing all of them, but if we did try to fix all of them we would outgrow it when we inevitably learn this material in later levels and the repair would essentially be the same measure theory we are about to do just with different language.

3.2 Riemann integration background

We define the Riemann integral for bounded functions on bounded domains as follows.

Let $f : [a,b] \to \mathbb{R}$ be bounded, then we say that $x_0 = a < x_1 < x_2 < ... < x_{n-1} < x_n = b$ is a partition of the interval $[a,b]$. For a given partition P We define a lower Darboux sum as

$$L(P) = \sum_{i=1}^{n} (x_i - x_{i-1}) \inf_{x \in [x_{i-1}, x_i]} f(x)$$

We define the upper Darboux sum as

$$U(P) = \sum_{i=1}^{n} (x_i - x_{i-1}) \sup_{x \in [x_{i-1}, x_i]} f(x)$$

We define the lower integral as the least upper bound of the Lower Darboux sums. We write this as

$$\underline{\int_a^b} f(x)\,dx = \sup_P L(P)$$

We define the upper integral by

$$\overline{\int_a^b} f(x)\,dx = \inf_P U(P)$$

We say f is Riemann integrable if the lower and upper integrals are equal. An equivalent condition is that for all $\varepsilon > 0$ there exists a partition P such that $U(P) - L(P) < \varepsilon$.

In 2D, we can define the integral of a bounded function on a closed rectangle by partitioning the x-coordinates and y-coordinates of the rectangle separately and defining your Darboux sums on the grid and proceeding similarly. We can do the same in higher dimensions.

We know that continuous functions are integrable. The same argument actually applies to 2D and 3D Riemann integrals as well to show the same result. There is actually a stronger result that characterizes which functions are integrable that we will prove later.

3.3 Measures

3.3.1 Definitions

Definition. Let X be any set, then $P(X)$ or the power set of X is the set of all subsets of X. For example, the power set of $\{1,2,3\}$ is

$$\{\emptyset, \{1\}, \{2\}, \{3\}, \{1,2\}, \{1,3\}, \{2,3\}, \{1,2,3\}\}$$

Definition. Let A, B be sets such that $A \subseteq B$, then $B \setminus A$ means the set of everything in B that is not in A. Actually we can drop the $A \subseteq B$ assumption and say $B \setminus A = B \cap A^C$, where A^C is shorthand for stuff not in A, or the complement of A.

Definition. A collection of sets is said to be disjoint if the intersection of any 2 of them is empty.

Definition. Given a collection of sets $A_1, A_2, A_3, ...$ we define $\prod_i A_i$ to be the set of all ordered lists $(a_1, a_2, ...)$ where $a_i \in A_i$ for each i. However, i does not have to range over the natural numbers, i can range over any set. Also, a common notation used is that the function π_j is a projection map i.e. it is a function from $\prod_i A_i$ to A_j that sends such an ordered list to the element a_j.

Definition. $\mathbb{N}$ is the set of positive integers. $\mathbb{Z}$ is the set of integers. $\mathbb{Q}$ is the set of rational numbers. $\mathbb{R}$ is the set of real numbers. $\mathbb{C}$ is the set of complex numbers.

Definition. Let $f : A \to B$ be a function i.e. a mapping between sets and let a be a subset of A and b be a subset of B. Then the image of a under f, sometimes written $f(a)$, is the set consisting of all $f(c)$ where c is an element of a. The pre-image of b under f, sometimes written $f^{-1}(b)$ is the set of all c in A such that $f(c)$ is in b.

Notation: We will write $f_n \nearrow f$ to mean that f_n is an increasing sequence converging to f, and $f_n \searrow f$ means the analogous thing.

Definition. Let X be a set. Then a σ-algebra $\mathcal{E}$ on X is a collection of subsets of X such that

1. The empty set is in $\mathcal{E}$

2. $A \in \mathcal{E}$ implies that $A^C = X \setminus A \in \mathcal{E}$

3. The union of any countable subset of $\mathcal{E}$ is in $\mathcal{E}$

We call the pair $(X, \mathcal{E})$ a measurable space.

Proposition. σ-algebras are closed under countable intersections. To see this, note that $\cap A_n = \left(\cup A_n^C\right)^C$

Note that if A, B are in a σ-algebra and $A \subseteq B$ then $A \cap B^C = A \setminus B$ is in the σ-algebra as well.

Definition. A measure on a measurable space is a function $\mu : \mathcal{E} \to [0, \infty]$ such that

1. The measure of the empty set is 0

2. Let A_n be a disjoint sequence of subsets of X that are in $\mathcal{E}$, then we have

$$\mu\left(\bigcup_n A_n\right) = \sum_{n=1}^{\infty} \mu(A_n)$$

Example. If E is a countable set, we can just define a "mass" m for each element and have the σ-algebra be the power set of E and define a measure by $\mu(A) = \sum_{x \in A} m(x)$ where A is any subset of E.

Note that if $m(x) = 1$ for all x we get the *counting measure.*

Definition. Let E be a set and A be a collection of subsets of E. Then we say A is a generator of a σ-algebra if the σ-algebra is

$$\{B \subseteq E : \text{B is in all } \sigma - \text{algebras that contain A}\}$$

We write $\sigma(A)$ to mean the σ-algebra generated by A. It is easy to check that this is indeed a σ-algebra as it contains everything it needs to contain to be one.

Example. If E is any countable set and A contains all sets of 1 element of E then $\sigma(A) = E$ since every subset can be written as a countable union of singletons.

Example. If $E = \mathbb{Z}$ and A contains all elements of the form $\{x, x+1, x+2, ...\}$ then by considering $\{x+1, x+2, x+3, ...\}$, taking its complement to get $\{..., x-2, x-1, x\}$ then intersecting that with the original set we get that $\{x\}$ is in $\sigma(A)$ for all x so $\sigma(A) = P(\mathbb{Z})$. Note that $P(S)$ is notation for the power set of S, or the set of all subsets of S.

Definition. The Borel σ-algebra is the σ-algebra on $\mathbb{R}$ generated by the set of all open subsets of $\mathbb{R}$. This is not the power set of $\mathbb{R}$ and we will see later why.

Definition. Let A be a collection of subsets of E, the set we care about. A is said to be a π-system if $a, b \in A$ implies $a \cap b \in A$. i.e. A is closed under finite intersections.

Definition. We say A is a d-system if $E \in A$ and $a, b \in A$ and $a \subseteq b$ implies that $b \setminus a \in A$, and it is the case that for any sequence a_n in A such that $a_n \subseteq a_{n+1}$ for all n (ie, any "increasing" sequence), we have the property that $\cup A_n \in A$.

Proposition. This follows straight forwardly from the definitions. A collection A is a σ-algebra if and only if it is both a π-system and a d-system.

Definition. (Ring) A collection of subsets A is a ring on E if it contains the empty set and $a, b \in A$ implies $a \cup b \in A$, and $b \setminus a \in A$.

Definition. (Algebra) A collection of subsets A is an algebra on E if it contains the empty set, and $a, b \in A$ implies $a^C \in A$, and $a \cup b \in A$.

Essentially it is a σ-algebra but closed under finite unions and not necessarily countable unions.

Theorem. (Dynkin's lemma) Let A be a π-system, then any d-system that contains A contains $\sigma(A)$

Proof. Let D be the intersection of all d-systems containing A. We will show that D contains $\sigma(A)$. Since anything that is a π and d system is a σ-algebra it is enough to check that D is a π-system since it contains A.

Note that A is a π-system and $A \subseteq D$ so A contains the empty set and we just need to check closure under finite intersections.

Define

$$D' = \{b \in D : b \cap a \in D \text{ for all } a \in A\}$$

Note that since A is a π-system it is closed under finite intersections and therefore every element of A is contained in D'. Specifically, if $a' \in A$ then $a' \cap a \in A \subseteq D$ so a' must be in D'.

Note that clearly $E \in D'$ and that if $b_1, b_2 \in D'$ where $b_2 \subseteq b_1$ then for any $a \in A$ we have

$$a \cap (b_1 \setminus b_2) = (a \cap b_1) \setminus (a \cap b_2)$$

By definition of D' we know that $b_1 \cap a$ and $b_2 \cap a$ are elements of D. Since D is a d-system we know that $(a \cap b_1) \setminus (a \cap b_2) \in D$ so $a \cap (b_1 \setminus b_2) \in D$.

Thus, since a was arbitrary we have that $b_2 \setminus b_1 \in D'$.

Now suppose that (b_n) is an increasing sequence in D' with $b = \bigcup b_n$. Then for any $a \in A$ we have

$$b \cap a = \left(\bigcup b_n\right) \cap a = \bigcup (b_n \cap a) \in D$$

The last inclusion is true because D is a d-system and $(b_n \cap a) \in D$ because $b_n \in D'$.

It follows now that $b \in D'$. Therefore, D' is a d-system contained in D which also contains A. By definition of D, it is contained in D' so $D = D'$.

Now let

$$D'' = \{b \in D : b \cap a \in D \text{ for all } a \in D\}$$

Since $D' = D$ it follows that $A \subseteq D''$ and $D'' \subseteq D$.

Now by the same argument as before starting after we showed that $A \subseteq D'$ but with D replacing the role of A, D'' is a d-system contained in D and therefore $D = D''$.

Since D contains the empty set and $D'' = D$ implies it is closed under taking the intersection of 2 elements (specifically if $a' \in D$ then it must be the case that $a \cap a'$ is in D for every a in D by definition of D'' which is equal to D), it is a π-system as required.

□

Definition. Let A be a collection of subsets of E containing the empty set. Then a set function is any function $\mu : A \to [0, \infty]$ satisfying $\mu(\emptyset) = 0$.

Definition. Let A be a collection of subsets of E containing the empty set. Then an increasing set function is any set function satisfying that for any a, b in A with $a \subseteq b$ we have $\mu(a) \leq \mu(b)$.

Definition. Let A be as above. Then an additive set function is any set function satisfying that for any a, b in A with $a \cup b$ in A and $a \cap b$ empty we have $\mu(a \cup b) = \mu(a) + \mu(b)$.

Definition. Let A be as above. Then a countably additive set function is any set function satisfying that for any a_n in A with $\cup a_n$ in A and the intersection of any 2 elements of the sequence a_n empty we have $\mu(\cup a_n) = \sum \mu(a_n)$.

Definition. Let A be as above. Then a countably subadditive set function is any set function satisfying that for any a_n in A with $\cup a_n$ in A we have $\mu(\cup a_n) \leq \sum \mu(a_n)$.

3.3.2 The Caratheodory extension theorem

Theorem. (Caratheodory extension theorem) Let A be a ring on E and let μ be a countably additive set function on A. Then μ extends to a measure on the $\sigma(A)$.

Proof. **Step 1.** We define μ^* for arbitrary sets $B \subseteq E$ as

$$\inf\left\{\sum \mu(a_n) : (a_n) \in A \text{ and } B \subseteq \bigcup a_n\right\}$$

If no such covering exists, then the measure is defined to be infinite. This may happen if B contains an element not contained in any element of A, for example. It may also happen if there is no covering with a finite such sum.

This is called the outer measure. For example, if A contains all open intervals, this is saying "The infimum of the length of sets of countably many intervals that cover the set" which is the outer measure.

We now define the set M of $\mu^* -$ measurable sets as the set of all a such that it is true that for all $B \subseteq E$ we have

$$\mu^*(B) = \mu^*(B \cap a) + \mu^*\left(B \cap a^C\right)$$

Note that μ^* is defined for arbitrary sets and we just don't know if it is "well behaved".

Step 2. We show that μ is countably subadditive and increasing.

Set $b_n = a_n \setminus \bigcup_{k<n} a_k$ for $n \geq 2$ and $b_1 = a_1$. Then the b_n's are disjoint, $b_n \subseteq a_n$ and $\bigcup_n b_n = \bigcup_n a_n$

We note that if $a_1 \subseteq a_2$ then $\mu(a_2) = \mu(a_1) + \mu(a_2 \setminus a_1)$ by countable additivity and the fact that μ is defined on a ring, thus μ is increasing. It is countably subadditive because whenever $\cup a_n \in A$ we have

$$\sum \mu(a_n) \geq \sum \mu(b_n) = \mu(\cup a_n)$$

Step 3. We show that μ^* is countably subadditive.

Let B be an arbitrary subset of E and suppose that $B \subseteq \cup B_n$. Then we need to show that $\mu^*(B) \leq \sum_n \mu^*(B_n)$. We can assume that for each n, $\mu^*(B_n)$ is finite, otherwise the inequality is trivial. Now fix any $\varepsilon > 0$. Then by definition of the outer measure as the infimum we can find a sequence $B_{n,m}$ in A with the property that

$$B_n \subseteq \bigcup_m B(n, m)$$

and

$$\mu^*(B_n) + \frac{\varepsilon}{2^n} \geq \sum_m \mu(B_{n.m})$$

Then it follows that

$$B \subseteq \bigcup_n B_n \subseteq \bigcup_{n,m} B(n, m)$$

Thus we have

$$\mu^*(B) \leq \sum_{n,m} \mu(B_{n,m})$$

by definition of outer measure so

$$\mu^*(B) \leq \sum_n \mu^*(B_n) + \frac{\varepsilon}{2^n} = \varepsilon + \sum_n \mu^*(B_n)$$

Thus we are done because ε was arbitrary.

Step 4. We will show that $\mu^* = \mu$ on A.

Suppose that $a \in A$ and that a_n is a sequence in A such that $a \subseteq \cup a_n$. By step 2 we know that μ is countably subadditive and increasing therefore we have

$$\mu(a) \leq \sum_n \mu(a \cap a_n) \leq \sum_n \mu(a_n)$$

Note that we needed to show both steps since it is not necessarily true that $\cup a_n$ is in A. Now taking the infimum over all such sequences it follows that

$$\mu(a) \leq \mu^*(a)$$

Note that $\emptyset$ means the empty set. Since $a, \emptyset, \emptyset, ...$ is such a sequence that covers a we must have by definition of the outer measure that

$$\mu^*(a) \leq \mu(a)$$

Thus $\mu^*(a) = \mu(a)$ completing step 4.

Step 5. We will show that M, our set of μ^*-measurable sets, contains A. Specifically, we need to show that if $a \in A$ and $B \subseteq E$ then

$$\mu^*(B) = \mu^*(B \cap a) + \mu^*\left(B \cap a^C\right)$$

By countable subadditivity we note that trivially

$$\mu^*(B) \leq \mu^*(B \cap a) + \mu^*\left(B \cap a^C\right)$$

For the other inequality we note that it is trivial if $\mu^*(B)$ is infinite. If it is finite then by that assumption we know a covering of B by countably many stuff in A exists, then given $\varepsilon > 0$ we can find some sequence B_n in A such that $B \subseteq \cup B_n$, and by the fact that we are taking the infimum we can in particular find a sequence B_n such that

$$\mu^*(B) + \varepsilon \geq \sum_n \mu(B_n)$$

Then we have

$$B \cap a \subseteq \bigcup_n (B_n \cap a)$$

$$B \cap a^C \subseteq \bigcup_n \left(B_n \cap a^C\right)$$

We note that since A is a ring we have

$$B_n \cap a^C = B_n \setminus a \in A$$

Therefore by definition of μ^* we have

$$\mu^* (B \cap a) + \mu^* \left(B \cap a^C\right) \leq \sum_n \mu (B_n \cap a) + \mu \left(B_n \cap a^C\right) = \sum_n \mu (B_n)$$

since μ is countably additive. Therefore

$$\mu^* (B \cap a) + \mu^* \left(B \cap a^C\right) = \sum_n \mu (B_n) \leq \mu^* (B) + \varepsilon$$

Since ε was arbitrary we have both

$$\mu^* (B) \leq \mu^* (B \cap a) + \mu^* \left(B \cap a^C\right)$$

and

$$\mu^* (B) \geq \mu^* (B \cap a) + \mu^* \left(B \cap a^C\right)$$

Thus we have

$$\mu^* (B) = \mu^* (B \cap a) + \mu^* \left(B \cap a^C\right)$$

so $A \subseteq M$ as required.

Step 6. We will show that M is an algebra.

We check the requirements. We will first show that $E \in M$. This is true since we have

$$\mu^* (B) = \mu^* (B \cap E) + \mu^* \left(B \cap E^C\right)$$

for all $B \subseteq E$, since $\mu^* (B \cap E) = \mu^* (B)$ and $\mu^* \left(B \cap E^C\right) = \mu^* (\emptyset) = 0$.

Now note that if a is in M then for all B we have by definition

$$\mu^* (B) = \mu^* (B \cap a) + \mu^* \left(B \cap a^C\right)$$

This definition is symmetric in a and a^C so $a^C \in M$.

We will show that M is closed under intersection. Note that when we have complements, this is equivalent to being closed under union. Suppose that $a_1, a_2 \in M$ and $B \subseteq E$. Then

$$\begin{aligned}
\mu^* (B) &= \mu^* (B \cap a_1) + \mu^* \left(B \cap a_1^C\right) \\
&= \mu^* (B \cap a_1 \cap a_2) + \mu^* \left(B \cap a_1 \cap a_2^C\right) + \mu^* \left(B \cap a_1^C\right) \\
&= \mu^* (B \cap (a_1 \cap a_2)) + \mu^* \left(B \cap (a_1 \cap a_2)^C \cap a_1\right) + \mu^* \left(B \cap (a_1 \cap a_2)^C \cap a_1^C\right) \\
&= \mu^* (B \cap (a_1 \cap a_2)) + \mu^* \left(B \cap (a_1 \cap a_2)^C\right)
\end{aligned}$$

This establishes that $a_1 \cap a_2 \in M$ as required.

Step 7. We will show that M is a σ-algebra

Let a_n be a disjoint sequence in M with union a. Then we want to show that $a = \cup a_n \in M$.

When we do this we will know that M is closed under disjoint unions. Since it is an algebra, if we have any countable sequence of sets $a_1, a_2, ...$ then the sets $b_n = a_n \setminus \bigcup_{k<n} a_k$ are in M and disjoint and it will follow that their union which is $\cup a_n$ must also be in M. Therefore we may indeed assume that a_n is a disjoint sequence

Suppose that $B \subseteq E$, then we have

$$\mu^*(B) = \mu^*(B \cap a_1) + \mu^*\left(B \cap a_1^C\right)$$

And also

$$\mu^*(B) = \mu^*(B \cap a_1) + \mu^*\left(B \cap a_1^C \cap a_2\right) + \mu^*\left(B \cap a_1^C \cap a_2^C\right)$$

But by disjointness this is just

$$\mu^*(B) = \mu^*(B \cap a_1) + \mu^*(B \cap a_2) + \mu^*\left(B \cap a_1^C \cap a_2^C\right)$$

In general for the same reason we have

$$\mu^*(B) = \sum_{i=1}^{n} \mu^*(B \cap a_i) + \mu^*\left(B \cap a_1^C \cap a_2^C \cap ... \cap a_n^C\right)$$

$$= \sum_{i=1}^{n} \mu^*(B \cap a_i) + \mu^*\left(B \cap \left(\bigcup_{i=1}^{n} a_n\right)^C\right)$$

$$\geq \sum_{i=1}^{n} \mu^*(B \cap a_i) + \mu^*\left(B \cap a^C\right)$$

since μ^* is increasing immediately from the definition. Taking the limit in n we get

$$\mu^*(B) \geq \sum_{i=1}^{\infty} \mu^*(B \cap a_i) + \mu^*\left(B \cap a^C\right)$$

By the countable subadditivity of μ^* (step 3) we have

$$\left(\sum_{i=1}^{\infty} \mu^*(B \cap a_i)\right) \geq \mu^*(B \cap a)$$

Thus we obtain

$$\mu^*(B) \geq \mu^*(B \cap a) + \mu^*\left(B \cap a^C\right)$$

But countable subadditivity implies the inequality in the other direction. Therefore

$$\mu^*(B) = \mu^*(B \cap a) + \mu^*\left(B \cap a^C\right)$$

Therefore by definition $a \in M$ so M is a σ-algebra.

Step 8. We will show that μ^* is countably additive on M.

Note that in the chain of inequalities from earlier

$$\mu^*(B) \geq \sum_{i=1}^{\infty} \mu^*(B \cap a_i) + \mu^*\left(B \cap a^C\right) \geq \mu^*(B \cap a) + \mu^*\left(B \cap a^C\right)$$

they must both be equalities by step 7. Taking $B = a$ we get

$$\mu^*(a) \geq \sum_{i=1}^{\infty} \mu^*(a \cap a_i) + \mu^*\left(a \cap a^C\right) = \sum_{i=1}^{\infty} \mu^*(a \cap a_i)$$

Therefore μ^* is countably additive.

□

Note that if A is a σ-algebra we can write

$$\mu^*(B) = \inf\{\mu(a) : a \in A \text{ and } B \subseteq a\}$$

This is called the outer measure.

3.3.3 Construction of the Lebesgue measure

Note that under some conditions we get uniqueness.

Proposition. Let A be a π-system on E with $\sigma(A) = \mathcal{E}$ and let μ_1, μ_2 be measures on $(E, \mathcal{E})$ with $\mu_1(E) = \mu_2(E) < \infty$ and $\mu_1 = \mu_2$ on A. Then $\mu_1 = \mu_2$

Proof. Define

$$D := \{a \in E : \mu_1(a) = \mu_2(a)\}$$

It would be enough to show that D contains $\sigma(A)$. Since A is a π-system contained in D, by Dynkin's lemma it is enough to show that D is a d-system.

By assumption it is true that $E \in D$.

If $a, b \in D$ and $a \subseteq b$ then by the fact that μ_1, μ_2 are measures and therefore countably additive on what we know is a σ-algebra, we have

$$\mu_1(b) = \mu_1(a) + \mu_1(b \setminus a) < \infty$$
$$\mu_2(b) = \mu_2(a) + \mu_2(b \setminus a) < \infty$$

Since a, b are in D it means that $\mu_1 = \mu_2$ for a, b and thus also for $b \setminus a$ so therefore $b \setminus a$ is in D as well.

Finally, let a_n be an increasing sequence in D with union a. Then

$$\mu_1(a) = \lim_{n\to\infty} \mu_1(a_n) = \lim_{n\to\infty} \mu_2(a_n) = \mu_2(a)$$

Therefore by definition $a \in D$.

Therefore D is a d-system so we are done.

$\square$

Example. Let $E = \mathbb{Z}$ and μ_1 be the counting measure and $\mu_2 = 2\mu_1$. Let A be the set of all sets of the form $\{x, x+1, x+2, ...\}$, then this is a π-system with $\sigma(A) = E$. But then we have that $\mu_1 = \mu_2 = \infty$ on A but clearly not in general. This is because the assumption that $\mu_1(E) < \infty$ failed.

Definition. The Borel σ-algebra on $\mathbb{R}$ is defined to be the σ-algebra generated by all open intervals. We denote this by $\mathcal{B}$. Note that many generating sets we may later quote generate the same object and it should be easy to see why in all such cases.

Theorem. There exists a unique Borel measure μ on $\mathbb{R}$ with $\mu([a, b]) = b - a$. This is called the *Lebesgue measure.*

Proof. **Step 1.** We will first show that if such a measure exists then it is unique.

Note that the measure of a single point is 0 (by considering making b as close as we want to a) so replacing one of the endpoints from a [to a (won't change anything.

Suppose $\tilde{\mu}$ is another such measure. Since our measure is not finite we cannot directly apply the previous uniqueness theorem. For each $n \in \mathbb{Z}$ we define

$$\mu_n(A) = \mu(A \cap (n, n+1])$$
$$\tilde{\mu}_n(A) = \tilde{\mu}(A \cap (n, n+1])$$

Then these are finite measures on $\mathcal{B}$ which agree on the π-system consisting of the empty set and intervals of the form $(a, b]$ with $a < b$. Therefore the hypotheses of the previous theorem are satisfied so we conclude that for all n we have $\mu_n = \tilde{\mu}_n$.

But then we have

$$\mu(A) = \sum_{n\in\mathbb{Z}} \mu_n(A) = \sum_{n\in\mathbb{Z}} \tilde{\mu}_n(A) = \tilde{\mu}(A)$$

The first equality is by countable additivity and by the fact that μ is being assumed to be a measure.

We can use the Caratheodory extension theorem to show existence but we have to be careful because the set of all open sets is not a ring. However, it is easy to see that the collection A that we are about to define is indeed a ring on $\mathbb{R}$ as the symmetric difference or union gives another set of a similar form. A is the set of subsets of the form

$$(a_1, b_1] \cup (a_2, b_2] \cup ... \cup (a_n, b_n]$$

where n is finite and

$$a_1 < b_1 < a_2 < b_2 < a_3 < b_3 < ... < a_n < b_n$$

Note that n can be 0 in which case we get the empty set.

Note also that $\sigma(A) = \mathcal{B}$ because we can get an open interval and every open subset of $\mathbb{R}$ can be written as a countable union of open intervals, and then we need to get whatever is generated by the open sets in order to have a σ-algebra. Specifically given an open interval (a, b) to prove it is in $\sigma(A)$ we take a countable union of elements in A as follows:

$$\left(a, b - \frac{b-a}{2}\right] \cup \left(a, b - \frac{b-a}{2^2}\right] \cup \left(a, b - \frac{b-a}{2^3}\right] \cup \left(a, b - \frac{b-a}{2^4}\right] \cup ...$$

We then set μ on A to be defined as $b_1 - a_1 + b_2 - a_2 + ... + b_n - a_n$. Then clearly it is additive if we have finitely many disjoint sets.

Step 2. Note that μ is clearly finitely additive. However, we need to show that it is countably additive so that we indeed have a measure that we can then extend using the Caratheodory Extension Theorem.

Let a_n be a disjoint sequence with union a and suppose it is such that $a \in A$. Then we need to show that $\mu(a) = \sum_{n=1}^{\infty} \mu(a_n)$. By finite additivity we have

$$\begin{aligned}\mu(a) &= \mu(a_1) + \mu(a \setminus a_1) \\ &= \mu(a_1) + \mu(a_2) + \mu(a \setminus (a_1 \cup a_2)) \\ &= \sum_{i=1}^{n} \mu(a_i) + \mu\left(a \setminus \bigcup_{i=1}^{n} a_i\right)\end{aligned}$$

We then just need to show that the error term tends to 0.

We let $a \setminus \bigcup_{i=1}^{n} a_i = b_n$ and we will assume that $\mu(b_n)$ does not tend to 0 which would mean that there exists ε such that $\mu(b_n) \geq \varepsilon$ for all n. For each n we take $C_n \in A$ with the property that its closure, C_n, is a subset of b_n and $\mu(b_n \setminus C_n) < \frac{\varepsilon}{2^{n+1}}$ which is possible since each b_n is just a finite union of intervals. So we have

$$\begin{aligned}\mu(b_n) - \mu\left(\bigcap_{m=1}^{n} C_m\right) &= \mu\left(b_n \setminus \bigcap_{m=1}^{n} C_m\right) \\ &\leq \mu\left(\bigcup_{m=1}^{n} b_m \setminus C_m\right) \\ &\leq \sum_{m=1}^{n} \mu(b_m \setminus C_m) \\ &\leq \sum_{m=1}^{n} \frac{\varepsilon}{2^{m+1}} = \frac{\varepsilon}{2}\end{aligned}$$

On the other hand we know that $\mu(b_n) \geq \varepsilon$ so we must have

$$\mu\left(\bigcap_{m=1}^{n} C_m\right) \geq \frac{\varepsilon}{2}$$

for all n. Now let K_n be $\bigcap_{m=1}^n \overline{C_m}$ (vertical bar above means closure), then $\mu\left(\bigcap_{m=1}^n C_m\right) \geq \varepsilon$ so K_n is non-empty for all n and it is a nested sequence. Therefore the intersection of all K_n is non-empty. But remember that $\bigcap_{m=1}^n \overline{C_m} \subseteq \bigcap_{m=1}^n b_m$ by definition of C so we now have

$$\emptyset \neq \bigcap_{n=1}^{\infty} K_n \subseteq \bigcap_{n=1}^{\infty} b_n = \emptyset$$

This is a contradiction so therefore we indeed have $\mu(B_n) \to 0$.

□

Definition. A measure μ on a measure space is said to be σ-finite if there exists a sequence E_n in $\mathcal{E}$ such that its union is E and $\mu(E_n) < \infty$ for all n.

Example. The Lebesgue measure on $\mathbb{R}$ or counting measure on $\mathbb{Z}$ is σ-finite but because $\mathbb{R}$ is uncountable the counting measure on $\mathbb{R}$ is not.

Proposition. The Lebesgue measure is translation invariant, ie $\mu(A+x) = \mu(A)$ for all borel sets A and real numbers x.

Proof. Let $\mu_x(A) = \mu(A+x)$. Then μ_x is a measure on $\mathcal{B}$ and $\mu_x([a,b]) = b-a$ so by the uniqueness above it is the same Lebesgue measure.

□

Proposition. Let μ be any Borel measure on $\mathbb{R}$ that is translation invariant and satisfies $\mu([0,1]) = 1$, then it is the Lebesgue measure.

Remark. This will mean that when I shortly show a non-measurable set, that it is not just that we have not defined our measure well enough but that it is literally not possible to define a sensible measure for arbitrary subsets of $\mathbb{R}$, at least without sacrificing either countable additivity or something else that such a measure ought to have. For example we can define the outer measure (ie the inf of length of open interval coverings) but that is not in general countably additive which it ought to be (like think if we were weighing physical copies of the set it ought to be additive).

Proof. By the above uniqueness theorem we will show that any such measure must satisfy $\mu([a,b]) = b-a$ for any a,b.

Note that by translation invariance it is clear that it works if $b = a+1$.

By translation invariance the measure of any point is 0. This is because otherwise by the increasing property on $[0,1]$ we could exhibit countably infintely many points and if they each had non-zero measure it would cause $[0,1]$ to have infinite measure. Therefore, $\mu\left(\left[a, a+\frac{1}{n}\right]\right) = \mu\left(\left[a, a+\frac{1}{n}\right)\right)$ and it is $\frac{1}{n}$by translation invariance because because

$$\mu\left(\left[a, a+\frac{1}{n}\right)\right) + \mu\left(\left[a+\frac{1}{n}, a+\frac{2}{n}\right)\right) + ... + \mu\left(\left[a+\frac{n-1}{n}, a+1\right)\right)$$

$$= \mu([a, a+1)) = \mu([a, a+1]) = 1$$

Thus it works for any rational length interval.

By considering countable additivity on these rational intervals it is easy to see that we can extend it to real intervals so we are done. Specifically, by letting q_n be some increasing sequence of rational numbers tending to your real number r.

□

It turns out that although if A is the ring of finite unions of half-open intervals it is true that $\sigma(A) = \mathcal{B}$, it turns out that the set of μ^*-measurable sets from the Caratheodory extension theorem is larger than it. We call this set $\mathcal{M}$ the Lebesgue σ-algebra.

We will often use the Borel σ-algebra since it has a nice generating set despite the fact that it is not the largest σ-algebra that we can put a sensible measure on.

It turns out that $\mathcal{M}$ is the set of all sets of the form $A \cup N$ where A is Borel and N is an arbitrary subset of a measure 0 set. Clearly all such sets are in $\mathcal{M}$ as they have an outer measure and satisfy the criterion, but it remains to show that $\mathcal{M} \subseteq \{A \cup N\}$.

To do this, note that given any X in $\mathcal{M}$ with finite outer measure we can find a sequence of open sets $O_k \supseteq X$ such that $\mu(O_k) \leq \mu^*(X) + \frac{1}{k}$. If you take the intersection of all O_k and get G you get a Borel set G such that $X \subseteq G$ and $\mu^*(G \setminus X) = 0$. You can do a similar trick from the inside of O_1 with $O_1 \setminus X$ and eventually you will end up with a Borel set $F \subseteq X$ such that $\mu^*(X \setminus F) = 0$. Thus X is the union of a borel set and a set with outer measure 0.

This proves it for finite outer measure and so it proves it for the measure on $[-n, n]$. For each n, set $X_n = X \cap [-n, n]$ and choose F_n Borel in X_n with $\mu^*(X_n \setminus F_n) = 0$, then $\cup F_n$ is a borel set differing from X by a measure 0 set.

Proposition. Countable sets have Lebesgue measure 0.

Proof. Fix $\varepsilon > 0$, then we want to cover our countable set by intervals with total length $\leq \varepsilon$. To do this, list the set as $a_1, a_2, a_3, ...$ and cover each a_n by an interval of length $\frac{\varepsilon}{2^n}$, then the length of the union of these intervals is $\leq \varepsilon$ by countable subadditivity.

□

3.3.4 A non-measurable set

We will now construct a non-measurable set.

Example. For $x, y \in [0, 1)$ we say $x \sim y$ if $x - y$ is rational. We will imagine putting all of $[0, 1)$ into boxes where we put x, y into the same box if $x \sim y$. We will now pick 1 element from each box and put them into a set $V \subseteq [0, 1)$. In this construction V is called a Vitali set. We will now show that V is not Lebesgue measurable.

For each rational number $r \in [0, 1)$ we define

$$V_r = \{(v + r) \bmod 1, v \in V\}$$

By translation invariance (applied once to the points to the left of r and again to the points to the right, using additivity) we know that if V is measurable so is V_r and that $\mu(V_r) = \mu(V)$. By definition of V we know that each V_r is disjoint and that their union equals $[0, 1)$. Thus by countable additivity we have $1 = \sum_r \mu(V)$. But then this is a contradiction because if $\mu(V) = 0$ then $1 = 0$ but if $\mu(V) > 0$ then $1 = \infty$.

Remark. You may be interested to look into the Banach-Tarski paradox. Essentially this carefully uses non-measurable sets to take a sphere and split it into 5 pieces and rotate those pieces around and reassemble them into 2 identical spheres.

3.3.5 Probability measures and independence

Definition. Suppose now that $(E, \mathcal{E})$ is a measure space and that $\mu(E) = 1$, then we call μ a probability measure and $(E, \mathcal{E}, \mu)$ a probability space. However, by convention we write probability spaces as (Ω, F, P) instead.

We call Ω, F, P the sample space, events, and probability of events respectively.

Here is how we want to think of this: Ω can be thought of as a box that we are gonna pull something at random from. The thing we pull from the box is called the outcome. The elements of F are called the events and they are subsets of Ω. For example, if Ω is a deck of cards, then the elements of F might be events like "you get a 7" or "you get spades", and these correspond to subsets of Ω and have a fixed probability. Therefore, the probability of an outcome is the probability of the event containing only that outcome, provided it is even in F in the first place.

We can interpret the Lebesgue measure on $[0, 1]$ as a probability measure. In this case, it becomes counterintuitive, as every outcome has probability 0 but yet we have to land on something. It is, therefore, a good thing that things have to be countably additive and not uncountably additive.

Definition. A sequence of events A_n is said to be independent if for all finite subsets $J \subseteq \mathbb{N}$ we have

$$P\left[\bigcap_{n \in J} A_n\right] = \prod_{n \in J} P\left(A_n\right)$$

Definition. A sequence of σ-algebras A_n where $A_n \subseteq F$ for all n is said to be independent if for every sequence a_n where $a_n \in A_n$ for all n we have that the sequence a_n is independent. It helps to think of these σ-algebras as sub-algebras of F.

Intuition for this definition. In a probability context, your σ-algebra tells you what questions you are allowed to ask about your outcome. If these are independent, it means that asking one question does not give you any information about asking a different question.

Proposition. Suppose $A_1$and A_2 are π-systems in F. If

$$P\left[a_1 \cap a_2\right] = P\left[a_1\right] P\left[a_2\right]$$

for all $a_1 \in A_1$ and $a_2 \in A_2$, then $\sigma\left(A_1\right)$ and $\sigma\left(A_2\right)$ are independent.

Proof. We will need the uniqueness theorem from earlier. As a reminder, this says that a finite measure is determined by its values on a π-system that generates the entire σ-algebra.

First fix $a_1 \in A_1$ and define the measures, for any $A \in F$ as

$$\mu\left(A\right) = P\left[A \cap a_1\right]$$

and

$$v\left(A\right) = P\left[A\right] P\left[a_1\right]$$

By assumption we know that these agree for $A \in A_2$ and we have that

$$\mu\left(\Omega\right) = P\left[a_1\right] = v\left(\Omega\right) \leq 1 < \infty$$

Since the above line is true and $\mu = v$ on a π-system A_2 it follows from the earlier uniqueness theorem that they agree on $\sigma\left(A_2\right)$. Thus for any $a_2 \in \sigma\left(A_2\right)$ we have

$$P\left[a_1 \cap a_2\right] = \mu\left(a_2\right) = v\left(a_2\right) = P\left[a_1\right] P\left[a_2\right]$$

By a symmetric argument with the roles of a_1 and a_2 reversed we know that the above equality is true for any $a_1 \in \sigma\left(A_1\right)$ as well.

Thus by definition of independence of σ-algebras the result follows.

$\square$

Corollary. Events A_n are independent if and only if the σ-algebras $\sigma\left(A_n\right)$ are independent.

Proof. It is clear that if the σ-algebras are independent then so are the events, by definition of independence.

To prove that the σ-algebras are independent we go by induction. Define our first π-system $\mathcal{A}_1$ as $\{A_1, \emptyset\}$ and our second π-system $\mathcal{A}_2$ as the collection consisting of the empty set and all finite intersections of the events $A_2, A_3, ..., A_n$. By independence of A_n we know that for any $A \in \mathcal{A}_1$ and $B \in \mathcal{A}_2$ we have $P\left[A \cap B\right] = P\left[A\right] P\left[B\right]$. By the previous theorem it follows that $\sigma\left(\mathcal{A}_1\right)$ and $\sigma\left(\mathcal{A}_2\right)$ are independent. Also, from this it follows that we can safely swap any event for its complement and we can do this not only for A_1 but for all n events.

To see why this implies mutual independence, consider as a concrete example the event $\left\{A_1^C, A_2, A_4, A_5\right\}$. Then

$$P\left[A_1^C \cap A_2 \cap A_4 \cap A_5\right] = P\left[A_1^C\right] P\left[A_2 \cap A_4 \cap A_5\right]$$

$$= P\left[A_1^C\right] P\left[A_2 \cap A_4 \cap A_5\right] = P\left[A_1^C\right] P\left[A_2\right] P\left[A_4 \cap A_5\right] = P\left[A_1^C\right] P\left[A_2\right] P\left[A_4\right] P\left[A_5\right]$$

Since this would work for any sequence of events in $\sigma\left(A_n\right)$ the result follows.

$\square$

3.4 Measurable functions

3.4.1 Definitions and basic properties

We would like to be able to talk about measurable functions - the idea is we want the "measure" of a function to be the "integral" or the "area under it". We will then get the Lebesgue integral and show that the result is actually much more general than the Riemann integral.

Definition. Let $(E, \mathcal{E})$ and $(G, \mathcal{G})$ be measure spaces. We say a function $f : E \to G$ is measurable if for every $A \in \mathcal{G}$ we have

$$f^{-1}(A) = \{x \in E : f(x) \in A\} \in \mathcal{E}$$

If $(G, \mathcal{G})$ is $([0, \infty], \mathcal{B})$ we say f is non-negative measurable.

As we will see though, when we actually measure it the codomain will typically be a subset of the reals or the complex numbers. As a convention we also allow our functions to take the value ∞.

The following proposition is useful for checking that a function is actually measurable.

Proposition. Let $(E, \mathcal{E})$ and $(G, \mathcal{G})$ be measure spaces and suppose that G is the σ-algebra generated by some family of sets $\mathcal{Q}$. Then if $f^{-1}(A) \in \mathcal{E}$ for every $A \in \mathcal{Q}$, then f is measurable.

Proof. We will claim that the set of A with the property that $f^{-1}(A) \in \mathcal{E}$ is actually a σ-algebra as then if it contains $\mathcal{Q}$ it must contain $\sigma(\mathcal{Q})$.

This is because it is closed under unions and complements and contains the empty set. For example, if some countable family A_n is in the set we are interested in, then so is $\cup A_n$ because

$$f^{-1}(\cup A_n) = \cup f^{-1}(A_n) \in \mathcal{E}$$

This is because $\mathcal{E}$ is known to be a σ-algebra.

Similarly f^{-1} preserves complements and emptiness and everything that a σ-algebra needs.

□

Remark. Note that if we have a function $f : E \to \mathbb{R}$ then to check it is measurable then by the proposition above we just need to check that $\{x \in E : f(x) \leq y\} \in \mathcal{E}$. The convention is that we are dealing with functions from $\mathbb{R}$ with the Lebesgue measure to $\mathbb{R}$ with the Borel measure. This is actually the least restrictive condition and allows the above criterion to be an exact criterion for measurability. However, because of the infinity condition we need to make sure that the set of points where f is infinite has measure 0 as well.

Example. We note that the indicator function of a set is a measurable function if and only if the set itself is measurable. This is immediate from the definition.

Example. The identity function (i.e. the function that sends every element to itself, $f(x) = x$, is always measurable.

Proposition. Let $(E, \mathcal{E})$, $(F, \mathcal{F})$ and $(G, \mathcal{G})$ be measurable spaces and let $f : E \to F$ and $g : F \to G$ be measurable functions, then the composition $g \circ f : E \to G$ is measurable.

Proof. We have that if $A \in \mathcal{G}$ then $g^{-1}(A) \in \mathcal{F}$ and thus $f^{-1}\left(g^{-1}(A)\right) \in \mathcal{E}$ by measurability of f and g and so $A \in \mathcal{G}$ implies $(g \circ f)^{-1}(A) \in \mathcal{E}$ so $g \circ f$ is measurable.

□

Definition. Suppose we have a set E and a family of real valued functions $\{f_i : i \in I\}$ on E, then we define

$$\sigma(f_i : i \in I) = \sigma\left(f_i^{-1}(A) : A \in \mathcal{B} : i \in I\right)$$

This is then the smallest σ-algebra on E which makes all the f_i's measurable.

Definition. Let $(E, \mathcal{E})$ and $(G, \mathcal{G})$ be measure spaces. We define the product measure space as $E \times G$ whose σ-algebra is generated by the sets $\{A \times B : A \in \mathcal{E}, B \in \mathcal{G}\}$. We can do the same for more than 2 measure spaces, we can form the product measure space of any family of measure spaces. However, it will be later that we see how to actually put a measure on these product spaces. If we have an infinite family, we have to be more careful. The product σ-algebra is actually given by the one above in this case.

Proposition. Let $f_i : E \to F_i$ be functions, then they are all measurable if and only if the function $g : E \to \prod F_i$ is measurable where the function g is defined by setting the ith component of $g(x)$ to be $f_i(x)$

Proof. If the map g is measurable then we know by composition with projection maps (i.e. maps that take a single component of an element of $\prod f_i$) that each f_i is measurable. This is because the pre-image of any element of the σ-algebra of f_i under projection is in the product measure space.

Conversely if each f_i is measurable then since the σ-algebra of $\prod f_i$ is generated by sets of the form $\pi_j^{-1}(A)$ where $A \in \mathcal{F}_j$, and by assumption the pre-image of these sets are in $\mathcal{E}$, the result follows by a theorem about needing to only check pre-images of sets in the generating set.

$\square$

Proposition. Let $(E, \mathcal{E})$ be a measure space and let f, g be measurable functions $E \to \mathbb{R}$, then $f + g$, fg, $\max(f, g)$ and $\min(f, g)$ are measurable.

Proof. These are all straight forward and go the same way. As an example, we can consider a function $E \to \mathbb{R}^2 \to \mathbb{R}$ that sends $x \to (f(x), g(x)) \to f(x) + g(x)$. Then by the proposition above the first function in the composition is measurable and by continuity (which we proved implies measurability) so is the second. Since we proved that compositions of measurable functions are measurable, the result follows. The exact same trick applies to the other 3 parts of the proposition so I leave them to you.

$\square$

Theorem. Let $(E, \mathcal{E})$ be a measure space and let $f_1, f_2, ...$ be a sequence of countably many measurable functions $E \to \mathbb{R}$, then $\inf_n f_n$, $\sup_n f_n$, $\liminf_n f_n$ and $\limsup_n f_n$ are all measurable if they give a non-infinite function.

Proof. For inf: If $f_n(x)$ is a sequence of measurable functions then $\inf(f_n)$ is a new function and the set of x for which $\inf(f_n) < a$ is exactly $\bigcup_{n=1}^{\infty} \{x : f_n(x) < a\}$ which is indeed a measurable set.

For sup: Exactly the same idea.

For liminf and limsup: Note that $\limsup(f_n) = \inf_{k \geq 1}\left(\sup_{n \geq k} f_n\right)$ which is measurable by the above, assuming the result is a finite function in which case this will be the case for all 'intermediate' functions in the composition.

$\square$

Notation. From now on we will write $f \wedge g$ for $\min(f, g)$ and $f \vee g$ for $\max(f, g)$

Theorem. (Monotone class theorem) Let $(E, \mathcal{E})$ be a measure space and let $\mathcal{A} \subseteq \mathcal{E}$ be a π-system with $\sigma(\mathcal{A}) = \mathcal{E}$, Let V be a set of functions $E \to \mathbb{R}$ such that

i) The constant function that sends everything in E to 1 is in V

ii) For all $A \in \mathcal{A}$ its indicator function, written as 1_A, is in V

iii) If f_n is a non-negative increasing sequence in V and its limit is a bounded function then its limit is also in V

iv) If $f, g \in V$ then $f + g \in V$ and for all $c \in \mathbb{R}$, $cf \in V$.

Then the claim is that V contains all bounded measurable functions.

Proof. Let $D = \{A \in \mathcal{E} : 1_A \in V\}$ is equal to $\mathcal{E}$. To do this we have to show that D is a d-system by Dynkin's lemma to show that D contains $\mathcal{E}$. But D is contained in $\mathcal{E}$ by its literal definition so therefore if we can show that D is a d-system then $D = \mathcal{E}$.

To show that D is a d-system we check the properties:

1. $1_E \in V$ so $E \in D$

2. if $1_A \in V$ and $A \subseteq B$ and $A, B \in D$ then we use the fact that V has $f, g \in V$ implies $cf \in V$ and $f + g \in V$ to say that $1_B - 1_A = 1_{B \setminus A} \in V$ so $B \setminus A \in D$

3. If A_n is an increasing sequence of sets in D then $1_{A_n} \to 1_{\cup A_n}$ and it is an increasing sequence so by definition we have $\cup A_n \in D$ so D is a d-system.

This means V contains indicators of all measurable sets.

Now suppose that f is bounded and non-negative measurable, then we want to show that $f \in V$. To do this we can approximate it by setting

$$f_n = 2^{-n} \lfloor 2^n f \rfloor$$

Note that $\lfloor x \rfloor$, often written floor (x), is defined to be the largest integer $\leq x$. Therefore,

$$f_n = \sum_{k=1}^{\infty} 2^{-n} 1_{\{f \geq k*2^{-n}\}}$$

Since f is bounded this is a finite sum that becomes 0 after some term, therefore it is a finite linear combination of indicators of elements in $\mathcal{E}$, and thus $f_n \in V$. But then we know that $0 \leq f_n \nearrow f$ so $f \in V$.

If f is bounded and not necessarily non-negative then write $f = (f \wedge 0) + (f \vee 0) = f^+ - f^-$, which by a previous proposition is the difference of 2 non-negative measurable functions, and therefore is also in V.

□

We will use this theorem when we do integration.

3.4.2 Constructing new measures

Definition. (Image measure) Let $(E, \mathcal{E})$ and $(G, \mathcal{G})$ be measure spaces. Suppose that μ is a measure on $\mathcal{E}$ and that $f : E \to G$ is a measurable function. We define the image measure ν on G for $A \in \mathcal{G}$ by $\nu(A) = \mu\left(f^{-1}(A)\right)$

It is easy to check that this is countably additive, since if A, B are disjoint so are their pre-images and we leverage countable additivity of μ, so ν is actually a measure.

We know that if we have a strictly increasing continuous function then it is invertible (if we restrict the codomain appropriately) and that the inverse is also strictly increasing[1]. It is also clear that these conditions (well it also works for a decreasing function) are necessary for an inverse of a continuous function to exist. However, if our function is increasing (not necessarily strictly) then we can get a 'generalized inverse' as we will see shortly.

A function g is said to be right continuous at x if it is 'continuous from the right', more formally that means that $x_n \searrow x$ implies $g(x_n) \to g(x)$. This is different from continuity because it only applies to decreasing sequences. As we will see in level 6.3, a cdf of a distribution should be right continuous.

[1] Well technically I don't think I've said this explicitly in any previous levels but it is pretty obvious since reflecting the graph about the line y=x doesn't change the fact that you can draw it without lifting your pen. I do the formal proof in the Analysis I undergraduate sublevel.

Lemma. Let $g : \mathbb{R} \to \mathbb{R}$ be non-constant, non-decreasing and right continuous. We define $g(\infty)$ to be $\lim_{n\to\infty} g(n)$ if this limit is finite or ∞ otherwise, and we define $g(-\infty)$ similarly. We then set I to be the open interval $(g(-\infty), g(\infty))$. This is non-empty because g is non-constant.

The lemma claims that the function $f : I \to \mathbb{R}$ defined by $f(x) = \inf\{y \in \mathbb{R} : x \leq g(y)\}$ is non-decreasing and left continuous and has the property that for all $x \in I$ and $y \in \mathbb{R}$ we have the following:

$$x \leq g(y) \Leftrightarrow f(x) \leq y$$

or equivalently

$$x > g(y) \Leftrightarrow f(x) > y$$

Proof. For $x \in I$ define $J_x = \{y \in \mathbb{R} : x \leq g(y)\}$. Then since g is non-decreasing we know that if $y \in J_x$ and $y' \geq y$ then $y' \in J_x$. By right continuity of g, if $y_n \in J_x$ and $y_n \searrow y$ then $y \in J_x$ and so therefore J_x is some interval $[L, \infty)$.

Note that $L = f(x)$ by the definition of f so we have the 2 implications from the lemma. It remains to show that f is left continuous and non-decreasing.

Clearly, $x \leq x'$ implies $J_{x'} \subseteq J_x$ which implies that f is non-decreasing.

Now if $x_n \nearrow x$ then $J_x = \bigcap_n J_{x_n}$ so $f(x_n) \to f(x)$ so f is left continuous.

□

Example. If figure 1 is g then figure 2 would be f. One can check from the definition that this is indeed the case.

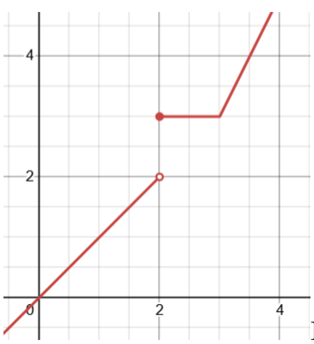


Figure 1

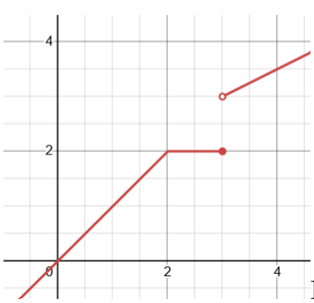


Figure 2

We will now use this to construct new measures on $\mathbb{R}$.

Let g be increasing and let f be its generalized inverse, then we can define the image measure as in the following Theorem.

Theorem. Let $g : \mathbb{R} \to \mathbb{R}$ be non-constant, non-decreasing and right continuous. Then there exists a unique measure dg on $\mathcal{B}$ such that $dg((a, b]) = g(b) - g(a)$.

Proof. Take I and f as in the previous lemma and let μ be the restriction of the Lebesgue measure to Borel subsets of I. Now f is measurable since it is left continuous (so pre-images of intervals, the generating set, are intervals). Now we define dg such that

$$dg\left((a,b]\right) = \mu\left(\{x \in I : a < f(x) \leq b\}\right) = \mu\left(\{x \in I : g(a) < x \leq g(b)\}\right) = g(b) - g(a)$$

The Caratheodory extension theorem extends this to a measure on all Borel sets.

The same proof that showed that the Lebesgue measure is unique works to show that dg is unique.

$\square$

Remark. This is useful because if g is a cumulative distribution function this gives us a probability measure on $\mathbb{R}$.

3.4.3 Random variables

Let (Ω, F, P) be a probability space and $(E, \mathcal{E})$ be a measurable space. Then we can define an E-valued random variable by any measurable function $X : \Omega \to E$.

This seems like a rather weird definition.

Example. We could make the random variable "1 if a coin is heads 0 otherwise" by letting X be the function $X : \Omega = (H, T) \to \mathbb{R}$ by $X(H) = 1$ and $X(T) = 0$.

Example. We could make a standard normal distribution by setting $\Omega = \mathbb{R}$, $F = \mathcal{B}$, and P defined by the measure above where our function g is given by the cdf of a standard normal. Then we would set $E = \mathbb{R}$ and X to be the identity function.

What we often do to encode a real random variable X is to set $\Omega = (0, 1)$ equipped with the Borel measure and then our function is the generalized inverse of the cdf.

Definition. (Distribution/Law) Given a random variable $X : \Omega \to E$ we say that the distribution or law of X is the image measure μ_X defined by

$$P(X \in A) = \mu_X(A) = P\left(X^{-1}(A)\right)$$

If $E = \mathbb{R}$ then we define

$$F_X(x) = \mu_X\left((-\infty, x]\right) = P(X \leq x)$$

This is the usual cumulative distribution function, and from properties of measures (i.e. continuity from below: That the measure of unions is the limit of the measure) we see that as $x \to \infty$ $F \to 1$ and that as $x \to -\infty$ $F \to 0$. F is also non-decreasing and right continuous and any such F with the right values at $\pm\infty$ gives a real random variable. With the same proof that we get a measure from such a function, such a function gives a random variable.

Note that we want to say that random variables X, Y are said to be independent if for any A, B in $\mathcal{E}$ (in fact any A, B in any π-system that generates $\mathcal{E}$) we have

$$P[X \in A, Y \in B] = P[X \in A]\, P[Y \in B]$$

But note that for each fixed X the events $X \in A$ for different subsets A form a π-system (easy to check) and similarly for Y and B.

We therefore define independence as follows:

Definition. A sequence of random variables X_n is said to be independent if the σ-algebras $\sigma(X_n)$ are independent.

We have now managed to generalize independence beyond the case where there is a density or the variables are discrete. We will prove several theorems that show that independence by this definition coincides with the properties we intuitively expect independence to satisfy.

Proposition. Two real-valued random variables X, Y are independent if and only if for any countable sequence of random variables $X_1, X_2, ...$ then they are independent if for every n and $x_1, x_2, x_3, ...$ and finite list of distinct natural numbers $a_1, a_2, ..., a_n$ we have

$$P\left[X_{a_1} \leq x_{a_1}, X_{a_2} \leq x_{a_2}, ..., X_{a_n} \leq x_{a_n}\right] = P\left[X_{a_1} \leq x_{a_1}\right] P\left[X_{a_2} \leq x_{a_2}\right] ...P\left[X_{a_n} \leq x_{a_n}\right]$$

Proof. The fact that independence of the σ-algebras implies the factorization is immediate from the definitions.

We need to show that if we have this factorization then the variables are independent. For that, we see that it is equivalent to the next theorem.

□

Theorem. (π-system lemma for independence) If $I_1, I_2, ..., I_n$ are independent π-systems then the σ-algebras they generate are also independent.

Proof. Fix arbitrary sets $A_2 \in I_2, A_3 \in I_3, ..., A_n \in I_n$ and define the following collection of sets:

$$D_1 = \{A \in \sigma\left(I_1\right) : P\left[A \cap A_2 \cap ... \cap A_n\right] = P\left[A\right] P\left[A_2\right] ...P\left[A_n\right]\}$$

Since the original π-systems are independent we know that $I_1 \subseteq D_1$.

To prove that D_1 contains $\sigma\left(I_1\right)$, by Dynkin's lemma we just need to show that D_1 is a d-system.

We know D_1 contains the empty set and the sample space it lives in. We will show that D_1 is closed under disjoint unions and set differences. Although this is slightly different from the actual definition of a d-system, it is easy to see that it is an equivalent definition.

For disjoint unions: Let E_k be a disjoint countable sequence in D_1 and E be their union. Then we have

$$P\left[E \cap A_2 \cap ... \cap A_n\right] = \sum P\left[E_k \cap A_2 \cap ... \cap A_n\right]$$

$$= \sum P\left[E_k\right] P\left[A_2\right] ...P\left[A_n\right] = P\left[E\right] P\left[A_2\right] ...P\left[A_n\right]$$

It therefore follows that $E \in D_1$. By a similar argument, we see that the set difference of 2 sets is in D_1 and therefore it is a d-system.

Repeating the process for I_2 then I_3 and so on, the result follows.

□

3.5 Integration

3.5.1 Lebesgue's criterion for Riemann integrability

We'll prove a useful theorem.

Theorem. Recall that a function is riemann integrable if overestimating or underestimating it with rectangles lets us get as close as we want and we proved that in 1D this is satisfied for continuous functions. Let Q be a rectangle in $\mathbb{R}^n$ and let $f : Q \to \mathbb{R}$ be a bounded function. Let D be the set of points of Q such that f is not continuous. Then if f is integrable over Q then D has measure 0.

It turns out that the converse holds as well: If D has measure 0 then f is Riemann integrable. However, we don't need the converse here and therefore we will prove it in a later level.

Proof. Define $B_\delta(a)$ to be the open set of points within δ of a.

Define the oscillation of f at a to be

$$\lim_{\delta\to 0}\left(\max_{\mathrm{B}_\delta(a)}(f) - \min_{\mathrm{B}_\delta(a)}(f)\right)$$

Then this is clearly 0 if and only if f is continuous by definition of continuity and definition of the limit.

Let D_m be the set of points where f has oscillation $\geq \frac{1}{m}$, then D is the union of all such D_m. Therefore if each D_m has measure 0 then we can cover it by open sets of total volume $\varepsilon 2^{-m}$ so that combining these covers gives a cover of D by open sets of total volume ε so we just need to show each D_m has measure 0.

Fix any m and $\varepsilon > 0$. Pick any partition P of Q for which the difference between its upper estimate and lower estimate for the integral is within $\frac{\varepsilon}{2m}$, possible as f is integrable. Let D'_m be the intersection of D_m with the boundary of P and $D''_m = D_m - D'_m$. D'_m has measure 0 as it is literally the boundary of rectangles so we can certainly cover it with total volume less than $\frac{\varepsilon}{2}$ by open sets. For D''_m let $R_{1,\dots,k}$ be the rectangles containing points of D''_m. Given i, R_i has a point in D''_m, and since this point (which we will call a) is not on the boundary of a rectangle there is a δ such that the cube c with side length 2δ centered around a is contained in R_i. Now by definition of oscillation

$$\frac{1}{m} \leq \max_c(f) - \min_c(f) \leq \max_{\mathrm{R_i}}(f) - \min_{R_i}(f)$$

On the other hand, by definition of Riemann integrability and the assumption about the partition P we have

$$\frac{\varepsilon}{2m} > \sum_{i=1}^{k}\left(\max_{R_i}(f) - \min_{R_i}(f)\right)\mathrm{vol}(R_i) \geq \sum_{i=1}^{k}\frac{\mathrm{vol}(R_i)}{m}$$

so those rectangles must have total volume less than $\frac{\varepsilon}{2}$.

∎

3.5.2 Definition of the Lebesgue integral

Now that we understand measurable functions we will try to understand how to actually measure them. The notation we will often use will be $\mu(f)$ or $\int_E f d\mu$. We by convention write dx instead of $d\mu$ if we are working with the Borel/Lebesgue measure on $\mathbb{R}$. Also, if μ is a probability measure then this integral can be written as $E[f(x)]$, ie the expectation of $f(x)$.

Note that Lebesgue measurable real functions are those that are measurable when we are going from $\mathbb{R}$ with the Lebesgue measure to $\mathbb{R}$ with the Borel measure. This allows indicator functions of non-Borel measurable sets to be measurable and makes it so we only have to check what happens on intervals to get our function to be measurable. Of all 4 combinations of Lebesgue/Borel in the domain and codomain this actually allows the *most* functions since it makes the pre-image of the "least" things allowed to be the "most" things, and it is easy to check as we only need to check intervals.

Definition. A simple function is a measurable function that can be written as a finite non-negative linear combination of indicator functions of measurable sets. Specifically, any simple function f is of the form

$$f = \sum_{k=1}^{n} a_k 1_{A_k}$$

for $A_k \in \mathcal{E}$ and $a_k > 0$.

Note that simple functions are exactly those that are measurable, non-negative and only take on finitely many values.

We will define the integral of a simple function defined as above to be $\sum_{k=1}^{n} a_k \mu(A_k)$

Note that this is well defined by basic properties of the measure. Well defined means that if different choices of A_k,a_k give the same simple function then the result will be the same.

For arbitrary non-negative measurable functions we define their integral as follows:

$$\mu(f) = \sup\{\mu(g) : g \leq f, g \text{ is simple}\}$$

Note the following remark: When we talk about assigning a measure to a function this is a function to the reals unless stated otherwise. It may also be to the *extended* reals where we allow $\pm\infty$ to be part of the codomain (note that $\mu(\infty) = 0$).

For arbitrary measurable functions f we find their integral by defining it to be $\mu(f \vee 0) - \mu((-f) \vee 0)$, i.e. considering the positive and negative parts. If we have an $\infty - \infty$ case we say this is undefined.

Definition. We say f is integrable if $\mu(|f|)$ is finite, which happens exactly when the integral of the positive and negative parts are finite.

If we are integrating over a subset of the reals then we call this the Lebesgue integral.

Theorem. Let $f : [a, b] \to \mathbb{R}$ be Riemann integrable. Then it is Lebesgue integrable and the integrals agree.

We will revisit this when we define a measure on $\mathbb{R}^n$ - it's not quite so simple as we will see.

Proof. **Step 1.** We will first prove that f is measurable. To do this we will use a theorem we proved earlier that says that if f is bounded on a bounded domain (standard conditions for Riemann integrability - and this theorem is not generally true for improper integrals, however it is true if the integral of the absolute value is finite, this is immediate by the dominated convergence theorem which we will soon see), then if it is Riemann integrable then it is discontinuous on a measure 0 set.

We will show that for any $U \subseteq \mathbb{R}$ its preimage is measurable. Let C be the set of continuity points and D the set of discontinuity points. Then

$$f^{-1}(U) = \left\{\left\{f^{-1}(U) \cap C\right\} \cup \left\{f^{-1}(U) \cap D\right\}\right\}$$

Note that $\left\{f^{-1}(U) \cap D\right\}$ has measure 0 so we just need to show that $\left\{f^{-1}(U) \cap C\right\}$ is measurable.

Let x be in $f^{-1}(U) \cap C$. Because U is open, there exists some $\varepsilon > 0$ such that $(f(x) - \varepsilon, f(x) + \varepsilon) \subseteq U$.

Clearly, $V \cap C$ contains all points in $f^{-1}(U) \cap C$ by definition. To see that the other side of the inclusion holds, note that V is the union of neighbourhoods which are all contained within $f^{-1}(U)$. Therefore, $V \subseteq f^{-1}(U)$, which immediately implies that $V \cap C \subseteq f^{-1}(U) \cap C$.

Since V is open, and C and $f^{-1}(U) \cap D$ are Lebesgue measurable we have

$$f^{-1}(U) = \left\{\left\{f^{-1}(U) \cap C\right\} \cup \left\{f^{-1}(U) \cap D\right\}\right\} = \left\{V \cup \left\{f^{-1}(U) \cap D\right\}\right\}$$

so $f^{-1}(U)$ is measurable.

Step 2. We assume for now that f is non-negative. We note that f is Riemann integrable hence bounded by some M, and so no simple function will have more measure then $M(b-a)$. Therefore the integral will be finite and still exist.

To show it equals the Riemann integral we use the fact that if $a \leq b$ are simple functions then $\mu(a) \leq \mu(b)$. This is pretty easy to show, by considering the simple function $b - a$ and noting that the integral of simple functions is additive.

Let the Riemann integral of f be R.

Now for every natural number n, we use Riemann integrability to make a partition and a rectangle-function overestimating f with integral between R and $R + \frac{1}{n}$. Notice, this is a simple function, and any simple function underestimating f will therefore be $\leq R + \frac{1}{n}$ by the fact above. On the other hand, the simple function that we get from Riemann-underestimating R by less than $\frac{1}{n}$ means that the supremum of the integrals of all such f-underestimating simple functions will be $\geq R - \frac{1}{n}$. Since this is true for all n we must have that the Lebesgue integral is R.

Step 3. To extend to arbitrary functions we simply use the positive part plus negative part characterisation of the Lebesgue integral and use linearity of the Riemann integral, then we're done. Note that if f is Riemann integrable then so is $\max(f, 0)$ since the difference between the upper and lower sums will always be strictly smaller.

$\square$

Example. Consider the restriction of $1_{\mathbb{Q}}$ to $[0,1]$. Then this is not Riemann integrable since any partition will have 1 as its upper estimate and 0 as its lower estimate. However, it is Lebesgue integrable as it is the indicator function of a measurable set. In fact its integral is 0.

3.5.3 Basic properties of the Lebesgue integral

In this subsubsection we will prove the Monotone convergence theorem which is a very important tool, as well as important properties of the Lebesgue integral such as linearity.

Lemma. This is obvious. Let f, g be non-negative measurable. Then $f \leq g$ implies $\mu(f) \leq \mu(g)$ (Monotonicity of Lebesgue integral)

Proof. Everything $\leq f$ is $\leq g$ so the sup of simple functions $\leq f$ is less than the sup of simple functions $\leq g$

□

Theorem. (Monotone convergence theorem) Suppose that (f_n) is a sequence of non-negative measurable functions and $f_n \nearrow f$. Then f is non-negative measurable and $\mu(f_n) \nearrow \mu(f)$.

Proof. **Step 1.** f is non-negative measurable because limits of measurable functions are measurable by an earlier theorem.

Step 2. Suppose f_n and f are indicator functions, so that $f_n = 1_{A_n}$and $f = 1_A$. Then $f_n \nearrow f$ and therefore $A_n \nearrow A$.

Using the fact that if $A_n \nearrow A$ then $A = \cup A_n$ and so $\mu(A) = \lim \mu(A_n)$, ie continuity of measures, which follows from countable additivity, we note that $\mu(A_n) \nearrow \mu(A)$ and therefore $\mu(f_n) \nearrow \mu(f)$.

Step 3. Suppose f is an indicator function. Suppose $f = 1_A$ for some fixed A. Fix $\varepsilon > 0$ and set $A_n = \{f_n > 1 - \varepsilon\}$ (which is in the σ-algebra by measurability of f_n). Then we know $A_n \nearrow A$ as every point in A_n eventually goes into A because $f_n \nearrow f$. Therefore we have

$$(1-\varepsilon) 1_{A_n} \leq f_n \leq f = 1_A$$

As $A_n \nearrow A$ we get

$$\lim_{n\to\infty} \mu(f_n) \geq (1-\varepsilon) \lim_{n\to\infty} \mu(A_n) = (1-\varepsilon)\mu\left(\lim_{n\to\infty} A_n\right) = (1-\varepsilon)\mu(f)$$

Since this is true for every ε the result follows.

Step 4. (General case) Let f be a general non-negative measurable function. Let s be any simple function such that $0 \leq s \leq f$, and fix a constant $c \in (0,1)$. Let $A_n = \{f_n \geq cs\}$.Because $f_n \nearrow f$ and $f \geq s$, for any point where $s(x) > 0$ we eventually have $f_n(x) \geq cs(x)$ as $n \to \infty$. Thus $A_n \nearrow \mathbb{R}$ (or the domain space).We know $f_n \geq f_n 1_{A_n} \geq cs1_{A_n}$. Because $s1_{A_n}$ is a simple function made of finitely many level sets, we can apply our result from Step 3 (continuity from below) and linearity over simple functions to take the limit as $n \to \infty$:

$$\lim_{n\to\infty} \int f_n d\mu \geq \lim_{n\to\infty} \int cs1_{A_n} d\mu = c \int s d\mu$$

Since this holds for any $c \in (0,1)$, letting $c \nearrow 1$ yields $\lim_{n\to\infty} \int f_n d\mu \geq \int s d\mu$. Finally, taking the supremum over all simple functions $s \leq f$ gives:

$$\lim_{n\to\infty} \int f_n d\mu \geq \int f d\mu$$

Combined with the trivial upper bound $\lim_{n\to\infty} \int f_n d\mu \leq \int f d\mu$ (since $f_n \leq f$), the result follows.

□

Theorem. We have the following properties

i) Let f, g be non-negative measurable functions between measure spaces. Then $\mu(af + bg) = a\mu(f) + b\mu(g)$ for $a, b \geq 0$ (Linearity of Lebesgue integral)

ii) Let f be non-negative measurable. Then $\mu(f) = 0$ if and only if $f = 0$ almost everywhere (i.e. except on a measure 0 set).

Proof. i) Let $f_n = 2^{-n}\lfloor 2^n f\rfloor \wedge n$, i.e. each value of f is cut off after its nth binary place. Let g_n be defined analogously. Then f_n and g_n are simple with $f_n \nearrow f$ and $g_n \nearrow g$ and therefore $af_n + bg_n \nearrow af + bg$. Thus by the monotone convergence theorem we have $\mu(af_n + bg_n) \nearrow \mu(af + bg)$.

Since f_n and g_n are simple functions we have

$$\mu(af_n + bg_n) = a\mu(f_n) + b\mu(g_n) \nearrow a\mu(f) + b\mu(g)$$

Therefore $\mu(af_n + bg_n)$ tends to both $\mu(af + bg)$ and $a\mu(f) + b\mu(g)$ so they are equal.

ii) Suppose f is not 0 almost everywhere and let $A_n = \left\{x : f(x) > \frac{1}{n}\right\}$. Then their union is the set of points where $f > 0$. Since this has positive measure it follows that some A_n has positive measure. Suppose its measure is ε, then $\mu(f)$ must be at least $\frac{\varepsilon}{n}$.

Conversely, if f is 0 almost everywhere then you cannot make an indicator function that covers a non-zero measure set so you will have to ask for the supremum of stuff that are all 0 to get $\mu(f)$.

□

Corollary. If g_n is a sequence of non-negative measurable functions then $\mu(\sum g_n) = \sum(\mu(g_n))$

Theorem. Monotonicity and linearity holds for arbitrary integrable functions, not just non-negative ones. Furthermore for arbitrary integrable f, if it is 0 almost everywhere then its measure is 0.

Proof. In what follows it is legitimate to decompose into positive and negative parts precisely because the functions are integrable.

The second part of the statement of the theorem is true with the same proof as in the previous theorem, therefore we just need to prove the first part.

By definition we have $\mu(-f) = -\mu(f)$. Given f, g let f^+ be $\max(f, 0)$ and f^- be $(-f)^+$ and let g^+ and g^- be defined similarly.

Suppose, then, that $a, b \geq 0$ since we can easily see by changing $g$to $-g$ if necessary that the result will follow from the restricted result to this assumption.

Also it is clear from the definition that $\mu(af) = a\mu(f)$. Therefore we just need to show that given f, g we have $\mu(f + g) = \mu(f) + \mu(g)$. We know this statement to be true for non-negative functions. Let $h = f + g$ and h^+ and h^- be defined analagously to f^+ and f^-. Then we have $h^+ - h^- = (f^+ - f^-) + (g^+ - g^-)$. We write

$$h^+ + f^- + g^- = h^- + f^+ + g^+$$

Now everything is non-negative measurable so we have linearity of μ here and can write

$$\mu(h^+) + \mu(f^-) + \mu(g^-) = \mu(h^-) + \mu(f^+) + \mu(g^+)$$

Rearranging gives

$$\mu(h^+) - \mu(h^-) = \mu(f^+) - \mu(f^-) + \mu(g^+) - \mu(g^-)$$

By definition of how we extend the measure to functions that are not necessarily non-negative the statement $\mu(h) = \mu(f) + \mu(g)$ follows so we are done.

Now we will prove monotonicity. Let $f \leq g$. Applying linearity and the fact that $\mu(g - f) \geq 0$, the result follows.

□

3.5.4 Limit theorems

Theorem. (Fatou's lemma) Let (f_n) be a sequence of non-negative measurable functions, then

$$\mu\left(\liminf\left(f_n\right)\right) \leq \liminf\left(\mu\left(f_n\right)\right)$$

Proof. Note that trivially if $k \geq n$ then

$$\inf_{m \geq n} f_m \leq f_k$$

So by monotonicity we have

$$\mu\left(\inf_{m \geq n} f_m\right) \leq \mu\left(f_k\right)$$

For all $k \geq n$. Taking infimum on both sides we have

$$\mu\left(\inf_{m \geq n} f_m\right) \leq \inf_{k \geq n} \mu\left(f_k\right)$$

Keeping the definition of lim inf in mind, note that by the monotone convergence theorem the left hand side converges to $\mu\left(\liminf\left(f_n\right)\right)$ and by the right hand side converges to $\liminf\left(\mu\left(f_n\right)\right)$. Then the result follows as limits preserve non-strict inequalities.

□

Remark. To keep the direction correct in your mind suppose $f_n = 1_{[n,n+1]}$ so that $\mu\left(\liminf\left(f_n\right)\right) = \mu\left(0\right) = 0$ but $\liminf\left(\mu\left(f_n\right)\right) = 1$.

The next result is the dominated convergence theorem. This result is incredibly useful. Finally all the 'Borel' and 'σ-algebra' nonsense that I kept seeing when trying to make A level maths rigorous and was frustrated I didn't know what it meant, is actually something that I understand now and it actually isn't so big and scary once I know what's going on.

Theorem. (Dominated convergence theorem) Let f_n be a sequence of measurable functions such that $f_n \to f$ pointwise almost everywhere and f is measurable. Suppose that there is an integrable function g with $|f_n| \leq g$ for all n. Then f is integrable and $\mu\left(f_n\right) \to \mu\left(f\right)$.

Note that this essentially says you can swap limits and integrals under such conditions.

Proof. Note that since $|f_n| \leq g$ for all n and limits preserve non-strict inequalities we know that $|f| \leq g$ so $\mu\left(f\right) < \infty$. From this and the fact that limits of measurable functions are measurable, we know f is integrable. However, it is only on a measure space where subsets of measure 0 sets are all measurable that a limit agreeing with f almost everywhere actually implies f is measurable.

Now we note that $g + f_n$ and $g - f_n$ are non-negative, measurable and integrable for all n.

We now will apply Fatou's lemma as follows:

$$\mu\left(g\right) - \mu\left(f\right) = \mu\left(g - f\right) = \mu\left(\liminf\left(g - f_n\right)\right) \leq \liminf\left(\mu\left(g - f_n\right)\right)$$

$$= \liminf\left(\mu\left(g\right) - \mu\left(f_n\right)\right) = \mu\left(g\right) + \liminf\left(-\mu\left(f_n\right)\right) = \mu\left(g\right) - \limsup\left(\mu\left(f_n\right)\right)$$

Since $\mu\left(g\right)$ is finite we can subtract it from all the equations to get

$$\mu\left(f\right) \geq \limsup\left(\mu\left(f_n\right)\right)$$

This is called the reverse Fatou lemma but it only necessarily holds when there is a dominating function g.

We will apply Fatou's lemma again to $g + f_n$ as follows:

$$\mu\left(g\right) + \mu\left(f\right) = \mu\left(g + f\right) = \mu\left(\liminf\left(g + f_n\right)\right) \leq \liminf\left(\mu\left(g + f_n\right)\right)$$

$$= \liminf\left(\mu\left(g\right) + \mu\left(f_n\right)\right) = \mu\left(g\right) + \liminf\left(\mu\left(f_n\right)\right) = \mu\left(g\right) + \liminf\left(\mu\left(f_n\right)\right)$$

Since $\mu\left(g\right)$ is finite we can subtract it from all the equations to get

$$\mu\left(f\right) \leq \liminf\left(\mu\left(f_n\right)\right)$$

These combine to tell us that

$$\mu\left(f\right) \geq \limsup\left(\mu\left(f_n\right)\right) \geq \liminf\left(\mu\left(f_n\right)\right) \geq \mu\left(f\right)$$

Thus these must all be equal to eachother, and so $\mu\left(f\right) = \lim\left(\mu\left(f_n\right)\right)$

Note that since the convergence is almost everywhere, the resulting error from it not being everywhere is 0.

□

We will now revisit the counterexample at the beginning of the section to show why domination by $g$is necessary. If $f_n(x)$ is defined as n for $0<x<1/n$ then the integral of this will be 1, but at all points between 0 and 1 f_n will eventually go to 0, so the integral of the limit is not the limit of the integral. It turns out that in this case we cannot find a function g such that for all n we have $f_n < g$ and $\int_0^1 g < \infty$ as the integral of any "bounding" function will be infinite if we try to compute it, since we can bound it below by $\sum_{n=1}^{\infty} \frac{1}{n}$, which as we saw in the series solutions proof in level 6.1 grows like $\ln\left(n\right)$.

3.5.5 Probability density functions

We can define new measures using a *density*. This can be a probability density but it doesn't have to. Let $(E, \mathcal{E}, \mu)$ be a measure space and let f be a non-negative measurable function. then we can define $\nu\left(A\right) = \mu\left(f 1_A\right)$ as a new measure on $(E, \mathcal{E})$

Proposition. ν as defined above is actually a measure.

Proof. Clearly it is 0 on the empty set. For countable additivity we note that if A_n is a disjoint sequence in $\mathcal{E}$ then

$$\nu\left(\cup A_n\right) = \nu\left(f 1_{\cup A_n}\right) = \mu\left(f \sum 1_{A_n}\right) = \sum \mu\left(f 1_{A_n}\right) = \sum \nu\left(A_n\right)$$

□

Note, also, that $\int (g)\, d\nu = \int (f \cdot g)\, d\mu$ for ν-integrable functions g, by taking indicator functions then simple functions and using the monotone convergence theorem and splitting into positive and negative parts.

Definition. A real random variable X has density $f_X \geq 0$ if $P\left(X \in A\right) = \int_A f_X\left(x\right) dx$ for every borel A. Necessarily, $\int_{\mathbb{R}} f_X\left(x\right) dx = 1$ and if f is piecewise continuous then it is the derivative of the cdf at all of the continuity points of f. Also, for every measurable function g for which the expectation is defined we have

$$E\left[g\left(X\right)\right] = \int_{\mathbb{R}} g\left(x\right) f_X\left(x\right) dx$$

Example. (Cantor distribution) The following random variable cannot be made as the sum of discrete and continuous parts:

Construct the cdf as follows:

X is always between 0 and 1 so we make the cdf 0 or 1 outside of this range. Then for $\frac{1}{3} \leq X \leq \frac{2}{3}$ we set the cdf to $\frac{1}{2}$. We now repeat the following procedure recursively to every interval where we have not defined X:

Take each interval where X has not been defined and take the middle third. Then set the cdf to be halfway between what it was in the last interval we defined X before and the first interval we defined X after. For example, for $\frac{1}{9} \leq X \leq \frac{2}{9}$

we would set the cdf to $\frac{1}{4}$ and for $\frac{7}{9} \leq X \leq \frac{8}{9}$ we set the cdf to $\frac{3}{4}$. We then cover every interval this way which will cover almost all X values and a dense set of cdf values, and we can fill in the cdf everywhere else to make the function continuous. Then it is not possible to meaningfully assign a density at any point since at any point in one of our intervals. Note: It is useful to know that what is left when removing the interior of all these middle third intervals is called the Cantor Set, known for being closed, containing no isolated points, having measure 0 and being uncountable. The reason we cannot make this random variable part discrete part continuous is because the cdf is continuous so it has no discrete atoms, and if we tried to assign a density it would have to integrate to 1 over a measure 0 set which is a contradiction.

3.6 Product measures

3.6.1 Definition and properties and iterated integrals

Given measure spaces E_1 and E_2 we want to assign a measure to $E_1 \times E_2$. Define the product σ-algebra as follows:

Definition. Let $\mathcal{E}_1$ and $\mathcal{E}_2$ be σ-algebras on E_1 and E_2 respectively. Then define

$$\mathcal{A} = \{A_1 \times A_2 : A_1 \in \mathcal{E}_1, A_2 \in \mathcal{E}_2\}$$

This is a π-system on $E_1 \times E_2$ and the product σ-algebra is $\sigma(\mathcal{A})$ which we write as $\mathcal{E}_1 \otimes \mathcal{E}_2$.

Let μ_1 and μ_2 be measures on $\mathcal{E}_1$ and $\mathcal{E}_2$ respectively. We would like to define the measure of $A \in \mathcal{E}_1 \otimes \mathcal{E}_2$ as

$$\int_{E_1} \left(\int_{E_2} 1_A(x_1, x_2)\, d\mu_2 \right) d\mu_1$$

where $x_1 \in E_1$ and $x_2 \in E_2$. However, we need to check that this definition makes sense.

Lemma. Let $E = E_1 \times E_2$ be a measurable space with the product σ-algebra $\mathcal{E}$ and let f be a $\mathcal{E}$-measurable function. Then

i) For each $x_1 \in E_1$ the function $x_2 \mapsto f(x_1, x_2)$ is $\mathcal{E}_2$-measurable

ii) Let E_2 be σ-finite. If f is a non-negative measurable then $f_1(x_1) = \int_{E_2} f(x_1, x_2)\, d\mu_2$ is $\mathcal{E}_1$-measurable.

Proof. Note that for each fixed x_1 the map $\pi : E_2 \to E$ given by $\pi(x_2) = (x_1, x_2)$ is measurable (since the pre-image of anything in the generating set of $\mathcal{E}$ is $\mathcal{E}_2$-measurable) and since f is measurable and compositions of measurable maps are measurable the result of part (i) follows.

For part (ii) we first prove the assertion if E_2 is finite so that we know that bounded measurable functions are integrable, then to extend to the σ-finite case we can use the monotone convergence theorem on an increasing sequence of finite measure sets tending to E_2 .

If we first assume f is bounded we can use the monotone class theorem. Let V be the set of all measurable functions such that $f_1(x_1) = \int_{E_2} f(x_1, x_2)\, d\mu_2$ is $\mathcal{E}_1$-measurable. Here note that we need to include ∞ in the codomain of f_1 since the integral is not necessarily finite. Then V satisfies $f, g \in V$ implies $cf \in V$ and $f + g \in V$ as the integral is linear. It is also clear that if $A \in \mathcal{A}$ where $\mathcal{A}$ is as in the definition of the product σ-algebra then $1_A \in V$ and $1_E \in V$. Therefore we have the generating π-system condition and just need to check closure under monotone limits.

Let (f_n) be a non-negative sequence in V such that $f_n \nearrow f$ and f is bounded. Then by the monotone convergence theorem we have

$$\left(\int_{E_2} f_n(x_1, x_2)\, d\mu_2 \right) \nearrow \left(\int_{E_2} f(x_1, x_2)\, d\mu_2 \right)$$

Thus the function sending x_1 to $\int_{E_2} f(x_1, x_2)\, d\mu_2$ is a limit of measurable functions and therefore measurable. Therefore we conclude by the monotone class theorem.

If f is not bounded then by considering $f \wedge n$ and taking the limit in n we note that f is in V as well.

$\square$

Theorem. Suppose E_1 and E_2 are σ-finite measure spaces with measures μ_1 and μ_2 respectively on σ-algebras $\mathcal{E}_1$ and $\mathcal{E}_2$ and that $E_1 \times E_2 = E$ with the product σ-algebra. There exists a unique measure μ on $\mathcal{E}$ such that $\mu(A_1 \times A_2) = \mu_1(A_1)\mu_2(A_2)$ for all $A_1 \in \mathcal{E}_1, A_2 \in \mathcal{E}_2$

Proof. By σ-finiteness we can take a sequence of subsets $E_{1,n}$ and $E_{2,n}$ such that they are finite measure spaces and their unions are E_1 and E_2. We can then restrict to these subspaces $E_{1,n} \times E_{2,n}$, prove uniqueness there, and take a union/limit of an increasing sequence of finite measure sets tending to E_1 and E_2 respectively to get that uniqueness holds everywhere, since products preserve limits. Therefore, even if I've explained it terribly, we can reduce to the case where we assume E_1 and E_2 are finite.

We will now try to define $\mu(A)$ as

$$\int_{E_1}\left(\int_{E_2} 1_A(x_1, x_2)\, d\mu_2\right) d\mu_1$$

Note that this is well defined by the previous lemma, actually a measure since it is countably additive by the monotone convergence theorem and is 0 on the empty set, and we have a working measure satisfying the product relation.

Uniqueness follows by a theorem we proved earlier that if measures agree on E and a generating π-system for the σ-algebra in question and E is finite then they are equal.

□

Example. We can get non-uniqueness by taking the product of $[0,1]$ with counting measure and $[0,1]$ with Lebesgue measure.

Hint. Try making one measure by summing the integrals of a set A restricted to $y = k$ over all k. Try making another one by integrating the number of elements in the set A restricted to $x = t$ over all t. Show that these both satisfy the product condition but give different measures for the set $(x, x) : x \in [0, 1]$.

Remark. Actually by the same argument as for what we did in $\mathbb{R}$ with the whole Caratheodory extension theorem outer measure stuff, in $\mathbb{R}^2$ any subset of a measure 0 set is measurable. However, the set of $(0, x)$ for x in a non-measurable subset of $\mathbb{R}$ is such a set and can therefore be sensibly assigned a measure in $\mathbb{R}^2$ despite not living in $\mathcal{M} \otimes \mathcal{M}$. To allow us to use properties of product measure in $\mathbb{R}^n$, we will assume that we are indeed working in $\mathcal{M} \otimes \mathcal{M} ... \otimes \mathcal{M}$ even when we could in theory use a larger σ-algebra.

Note that we could have also defined the measure by

$$\int_{E_2}\left(\int_{E_1} 1_A(x_1, x_2)\, d\mu_1\right) d\mu_2$$

But by uniqueness these are the same! We will show this in the following theorem.

Theorem. (Fubini/Tonelli's theorem) Let f be non-negative measurable on $E = E_1 \times E_2$ with respect to the product measure where E_1 and E_2 are both σ-finite measure spaces. Then

$$\int_{E_2}\left(\int_{E_1} f(x_1, x_2)\, d\mu_1\right) d\mu_2 = \mu(f) = \int_{E_1}\left(\int_{E_2} f(x_1, x_2)\, d\mu_2\right) d\mu_1$$

Proof. The equality is known for indicator functions of product-measurable sets. Extend it by finite non-negative linearity to simple functions. For arbitrary $f \geq 0$, choose simple $s_n \nearrow f$ and apply the monotone convergence theorem to the joint integral and both iterated integrals.

□

Theorem. Let f be a Riemann integrable bounded function from a closed rectangle in $\mathbb{R}^n$ to $\mathbb{R}$. Then there exists a measurable function g such that $g = f$ almost everywhere and $\int_{\text{Lebesgue}} g = \int_{\text{Riemann}} f$

Proof. By boundedness choose C with $f + C \geq 0$ to allow subsequent monotone convergence theorem steps. Alternatively, just use dominated convergence. Note that Riemann integrability makes the upper and lower integrals converge to the same value by definition.

Let P_n be a partition of mesh 2^{-n} such that P_{n+1} is always a refinement of P_n and let U_n and L_n be the best upper and lower step functions on P_n. Then $\int L_n \nearrow \int_{\text{Riemann}} f$ by Riemann integrability and therefore $\int_{\text{Riemann}} f = \int_{\text{Lebesgue}} \sup(L_n)$ by the monotone convergence theorem, as L_n is an increasing sequence and $\sup(L_n)$ is Borel measurable. We can do a similar argument to show that $\int_{\text{Riemann}} f = \int_{\text{Lebesgue}} \inf(U_n)$. Therefore $\int_{\text{Lebesgue}} (\inf(U_n) - \sup(L_n)) = 0$. Since $\inf(U_n) - \sup(L_n)$ is non-negative it must be 0 almost everywhere by a previous theorem about Lebesgue integrals. Thus $\sup(L_n)$ works as a g.

$\square$

This is the best we can do, the following example shows why.

Let $V \in [0,1]$ be non-measurable, then define $f : [0,1] \times [0,1] \to \mathbb{R}$ as follows:

$$f = \begin{cases} 0 & y > 0 \\ 0 & y = 0, x \notin V \\ 1 & y = 0, x \in V \end{cases}$$

Then by making the partition sufficiently fine in y the upper Riemann sum tends to 0 so f is Riemann integrable, but it is not Lebesgue integrable.

Proposition. Let E be as above and let f be integrable on E and let A be defined as

$$A = \left\{ x_1 \in E_1 : \int_{E_2} |f(x_1, x_2)| \, d\mu_2 < \infty \right\}$$

Then $\mu_1(E_1 \setminus A) = 0$. Also, if we set

$$f_1(x_1) = \begin{cases} \int_{E_2} f(x_1, x_2) \, d\mu_2 & x_1 \in A \\ 0 & \text{Otherwise} \end{cases}$$

then f_1 is a μ_1-integrable function and $\mu_1(f_1) = \mu(f)$

Proof. Since the function $x_1 \longmapsto \int_{E_2} |f(x_1, x_2)| \, d\mu_2$ is μ_1-integrable by μ-integrability of f, it can only be infinite on a measure 0 set.

Now define

$$f_1^+(x_1) = \int_{E_2} [f(x_1, x_2) \vee 0] \, d\mu_2$$

$$f_1^-(x_1) = -\int_{E_2} [f(x_1, x_2) \wedge 0] \, d\mu_2$$

Then by definition $f_1(x_1) = 1_A \left(f_1^+(x_1) - f_1^-(x_1) \right)$ and the result follows since

$$\mu(f) = \mu\left(f^+\right) - \mu\left(f^-\right) = \mu_1\left(f_1^+\right) - \mu_1\left(f_1^-\right)$$

and

$$|f_1(x_1)| \leq \int_{E_2} |f(x_1, x_2)| \, d\mu_2$$

by Fubini/Tonelli. Therefore, since it is ok for functions to differ on a measure 0 set and it does not affect the integral it follows that

$$\mu(f) = \mu_1(f_1)$$

$\square$

This means that as long as f is product measurable and the integral of its absolute value is finite, we can swap the order of integrals.

Theorem. Let $f : \mathbb{R} \to \mathbb{R}$ be non-negative measurable and integrable, then the area measure of the area under f equals $\mu(f)$

Proof. Define the area as follows:

$$A_f = \left\{(x,y) \in \mathbb{R}^2 : 0 \le y \le f(x)\right\}$$

The set is product measurable because $(x,y) \to f(x) - y$ ie measurable and it is the set where $y \ge 0$ and that function is ≤ 0. By the theorem above, we have

$$\mu(A_f) = \int_{\mathbb{R}} \left(\int_{\mathbb{R}} 1_{A_f}(x,y)\, dy\right) dx = \int_{\mathbb{R}} f(x)\, dx$$

□

3.6.2 Application to probability and independence

Proposition. Let $X_1, ..., X_n$ be random variables in $(\Omega, \mathcal{F}, P)$ with values in $(E_1, \mathcal{E}_1), ..., (E_n, \mathcal{E}_n)$ respectively. Then let $E = E_1 \times ... \times E_n$ with the product σ-algebra $\mathcal{E}$. Then $X = (X_1, X_2, ..., X_n)$ is $\mathcal{E}$-measurable

Proof. This follows as the generating set is just the sets of the form $A_1 \times A_2 \times ... \times A_n$ with $A_i \in \mathcal{E}_i$ for each i, and the pre-image of each of these generating sets is the intersection over all i from 1 to n of the pre-images of A_i, which is contained in $\mathcal{F}$ as $\mathcal{F}$ is a σ-algebra.

□

Proposition. Let $X_1, ..., X_n$ be as above. Then the following are equivalent:

i) They are independent

ii) $\mu_X = \bigotimes_{i=1}^n \mu_{X_i}$

iii) For all f bounded and measurable,

$$E\left[\prod_{k=1}^n f_k(X_k)\right] = \prod_{k=1}^n E\left[f_k(X_k)\right]$$

Proof. **Step 1.** (i) implies (ii)

Let $\nu = \bigotimes_{i=1}^n \mu_{X_i}$. Then we just need to check that ν and μ_X agree on the π-system generating the entire σ-algebra. Let

$$\mathcal{A} = \{A_1 \times ... \times A_n : \forall i A_i \in \mathcal{E}_i\}$$

Then for each $A \in \mathcal{A}$ we have, by independence,

$$\mu_X(A) = P[X \in A] = P\left[\bigcap X_i \in A_i\right] = \prod P[X_i \in A_i] = \nu(A)$$

Step 2. (ii) implies (iii)

We will use Fubini's theorem here. Note that

$$E\left[\prod_{k=1}^n f_k(X_k)\right] = \int_E \prod_{k=1}^n f_k(x_k)\, d\mu_X = \prod_{k=1}^n \int_{E_k} f_k(x_k)\, d\mu_{X_k} = \prod_{k=1}^n E\left[f_k(X_k)\right]$$

Step 3. (iii) implies (i)

Take $f_k = 1_{A_k}$ for $A_k \in \mathcal{E}_k$. Then we have

$$P\left[\bigcap X_i \in A_i\right] = E\left[\prod_{k=1}^{n} 1_{A_k}(X_k)\right] = \prod_{k=1}^{n} E\left[1_{A_k}(X_k)\right] = \prod_{k=1}^{n} P\left[X_i \in A_i\right]$$

Therefore the X_i's are independent.

□

4 Intuition for complex integration

Also, in my exponentials and logarithms video, I briefly mentioned that if a function has a single-valued antiderivative then the integral along a path of that function does not depend on the path. By "integrate along a path", I mean take the sum of the function times the complex displacement you move by on the path (eg if you move from $1.07 + 3.12i$ to $1.08 + 3.14i$ you add $(0.01 + 0.02i)f(1.07 + 3.12i)$), then take the limit of these sums as these distances go to 0. Intuitively, similar to actual integration, taking the sum like this moves you along the antiderivative, which if it is single valued (unlike log, for example) will not allow you to possibly have path dependence.

Note: We will formalize this in the complex analysis undergraduate sublevel. For now this tells you why it is true.

5 Uniform continuity

Definition. A function is said to be uniformly continuous if given an $\varepsilon > 0$ in the definition of continuity not only are we able to pick a δ for each x, but a δ that works for all x. An example of a function that is not uniformly continuous is 1/x on the open interval (0,1), since if I pick a certain ε, then no matter how small I pick δ, I can go close enough to 0 that the $|x - a| < \delta$ implies $|f(x) - f(a)| < \varepsilon$ condition is not satisfied, so I cannot pick a δ that works for all x.

Specifically,

Continuous: A function is continuous if for all x in its domain it is true that for all $\varepsilon > 0$ there is a $\delta > 0$ such that $|x - y| < \delta \implies |f(x) - f(y)| < \varepsilon$

Uniformly continuous: A function is uniformly continuous if for all $\varepsilon > 0$ there is a $\delta > 0$ such that for all x in its domain it is true that$|x - y| < \delta \implies |f(x) - f(y)| < \varepsilon$

This difference is subtle but it is important. For example, $\frac{1}{x}$ for $x > 0$ is continuous but not uniformly continuous, since for each ε if I pick a δ and go close enough to 0 it will stop being a δ that works. However, there is something useful we can say on the matter.

Theorem. Suppose that f is a function which is continuous on a closed bounded interval (or more generally, a closed bounded subset of $\mathbb{R}^n$), then it is uniformly continuous

Proof. Since we are assuming our function is continuous on a closed bounded set, the Bolzano Weierstrass theorem applies by boundedness, and the subsequence in question converges to a limit in the interval by closure of the interval.

We will see similarities to the level 4 proof that continuous functions are integrable.

Assume for a contradiction that our function is continuous but not uniformly continuous on our closed bounded interval. Then there is an $\varepsilon > 0$ such that if I pick $\delta = \frac{1}{n}$ there is some x_n, y_n with $|x_n - y_n| < \frac{1}{n}$ but $|f(x_n) - f(y_n)| > \varepsilon$. We know that x_n and y_n both have convergent subsequences converging to places on our closed interval, and in fact they converge to the same place since the difference between x_n and y_n gets arbitrarily small. If the subsequences in question are x_{n_i} and y_{n_i} and their limits are x, then $f(x_{n_i}) \to f(x)$ and $f(y_{n_i}) \to f(x)$ because continuous functions preserve limits. But $f(x_{n_i})$ and $f(y_{n_i})$ somehow converge to the same limit but are always $> \varepsilon$, which is a contradiction.

□

6 Calculus results

6.1 Big O and little o notation

Definition. A function $g(x)$ is said to be $O(f(x))$ if when x is sufficiently close to x_0, $|g(x)|$ is bounded by $M|f(x)|$ where M is some fixed positive constant. If $f(x)$ is not 0 in the vicinity of x_0 we can write that $|\frac{g(x)}{f(x)}| < M$ when x is sufficiently close to x_0. We can also say, and this is often done in computer science when analyzing how long an algorithm will take, that a function $g(x)$ is $O(f(x))$ as x goes to infinite if when x is large enough, $|g(x)|$ is bounded by $M|f(x)|$. The value of x_0 in a specific situation is often implied by context. Sometimes people write $g(x) = O(f(x))$

Definition. Little o notation is like big O notation except if $g(x)$ is $o(f(x))$ at x_0 it means that $g(x)$ gets much smaller than $f(x)$ when x is sufficiently close to x_0. Precisely, it means that M can be made arbitrarily small by making x sufficiently close to x_0, or by making x sufficiently large in the case x_0 is infinity.

Note that if $f(x) = o(g(x))$, then we necessarily also have $f(x) = O(g(x))$.

Example. $x \neq O\left(x^2\right)$ as $x \to 0$ since x^2 is much smaller than x. In fact, $x^2 = o\left(x\right)$ as $x \to 0$.

$x = O\left(x^2\right) = o\left(x^2\right)$ as $x \to \infty$ but $x^2 \neq O\left(x\right)$ as $x \to \infty$.

$x = O\left(\sqrt{x}\right)$ as $x \to 0$.

$\sin(2x) = O\left(x\right)$ as $x \to 0$, but $\sin(2x) \neq o\left(x\right)$ as $x \to 0$.

$x \neq O\left(x\sin(x)\right)$ as $x \to \infty$, because although x does not seem to be much larger than $x\sin(x)$, the ratio $\frac{x}{x\sin(x)}$ is unbounded when x gets close to integer multiples of π.

Also, a principle here is the idea that a polynomial is dominated by its leading term. For example

$2x^3 + 4x + 12 = O(x^3)$ as $x \to \infty$ because clearly eventually x^3 exceeds $4x + 12$ so the whole thing is less than $3x^3$ so we can put $M = 3$. Also, big o notation can capture the idea that exponentials beat polynomials (by the way you can prove this as $\frac{a^x}{x^b}$ goes to infinity as x goes to infinity which you can see if you apply l'hopital's rule at least b times), as we can write that $P(x) = o(\exp(x))$ for any polynomial P. Also, $\log\left(x\right) = o(x^\varepsilon)$ for all positive ε and this can be proven the same way, or by thinking of it intuitively as a sort of inverse of the exponentials beat polynomials idea. Note $\exp(x)$ means e^x.

Another idea is that non-zero constants don't matter here since we're looking at the bigger picture with the growth rate of these functions.

Example. Since $\lim_{h\to 0} \frac{f(x+h)-f(x)}{h} = f'(x)$, $\lim_{h\to 0} \frac{f(x+h)-f(x)}{h} - f'(x) = 0$ so $\frac{f(x+h)-f(x)}{h} - f'\left(x\right) = o(1)$ as $h \to 0$ by definition. Therefore, $f\left(x+h\right) - f(x) - hf'\left(x\right) = o(h)$ by multiplying by h on both sides, so we can write that $f\left(x+h\right) - f\left(x\right) = hf'\left(x\right) + o(h)$, which looks a lot like what I did when I justified differentiating infinite power series.

6.2 Taylor's theorem

Definition. A taylor polynomial of degree n is a taylor series for a function that is n times differentiable but only up to the term in x^n. We write this as $T_n(f)(x)$. This is a polynomial which gives an approximation of f near some point x_0, with error equal to $o((x-x_0)^n)$.

Claim. If f is n times differentiable at x_0 then $f\left(x\right) - T_n\left(f\right)\left(x\right) = o((x-x_0)^n)$ as $x \to x_0$ if x_0 is the point we are doing a taylor series around.

Claim. If f is n+1 times continuously differentiable on the interval $[x_0,\ x]$ then $f\left(x\right) - T_n\left(f\right)\left(x\right) = O((x-x_0)^{n+1})$ as $x \to x_0$ if x_0 is the point we are doing a taylor series around.

Claim. If the conditions for claim 2 hold, then there exists some $t \in (x_0,\ x)$ such that $f\left(x\right) - T_n\left(f\right)\left(x\right) = \frac{(x-x_0)^{n+1}}{(n+1)!} f^{(n+1)}(t)$.

We will prove this.

Lemma. (Extreme value theorem) Any function continuous on a finite closed interval is bounded and actually attains its minimum and maximum.

Proof. Note: It is very important that the interval is closed. On the open interval (0, 1) we can define 1/x which is continuous on that interval, but not on the closed interval [0,1] because the closed interval contains $x = 0$ but the open interval does not. But there's not much you can do to make a continuous unbounded function defined on a finite closed interval – Try it!

We will use the fact that every bounded sequence has a convergent subsequence since that was proven in level 4.

Suppose for a contradiction f is unbounded. Then for all n, we can choose x_n in [a,b] with $|f(x_n)| > n$. Now pick a convergent subsequence x_{n_k} of the sequence formed by each x_n, and call its limit L, which is in $[a, b]$ since $[a, b]$ is closed so limits of sequences in [a,b] are in [a,b]. By continuity, $f(x_{n_k}) \to f(L)$, but since $f(x_{n_k})$ is unbounded, it follows that this is a contradiction, since $f(L)$ cannot be defined.

Important note: Bolzano weierstrass holds for bounded sequences of points in k dimensions, and if the bounded set in question is closed then the limit of the subsequence will also be in the set – the reason is that there is a subsequence whose x coordinate converges and a subsequence of that whose y coordinate converges etc, and thus the above theorem holds for functions of more than one variable, as well as the theorem about uniform continuity and about Riemann integrals for continuous functions, as the definition of continuity is analogous. This is important since we use these theorems in the context of functions of multiple variables.

We will now prove that the maximum is attained. By boundedness there is a least upper bound M. Therefore there is a sequence of increasing points $M_1, M_2, \ldots$ converging to M that are values of the function. For each of these points pick an input value t_k of f such that $f(t_k) = M_k$. Now t_k has a convergent subsequence converging to t and by continuity $f(t) = M$, meaning the maximum value of a function on a closed interval is attained. It is by closedness that we can guarantee t is in our interval.

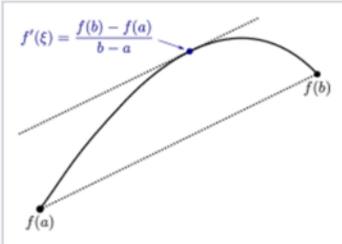
□

Lemma. (Mean value theorem)

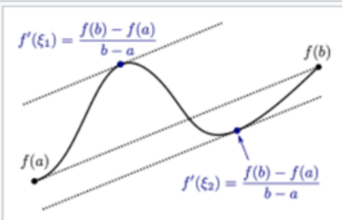
If we have a continuous function on [a,b] differentiable on the open interval (a,b) then at some point in (a,b) the derivative is equal to its mean value - specifically the slope of the line joining the endpoints.

This lemma assumes the values are real, so everything proven here applies to real numbers, but not general complex numbers.

Proof. First of all, a picture (Figure 3, from Wikipedia):



The function f attains the slope of the secant between a and b as the derivative at the point $\xi \in (a, b)$.



It is also possible that there are multiple tangents parallel to the secant.

Figure 3

You can see that it's kind of obvious that the function's derivative attains its mean value, since the function is differentiable everywhere by assumption so there are no spikes. This is essentially what the theorem says. However, it is not completely obvious as derivatives can be discontinuous.[2]

To prove this, we want to subtract the line from $(a, f(a))$ to $(b, f(b))$ from y. This line has the slope we want so we want to prove that our new function, which is 0 at a and b, attains a derivative of 0 between a and b. This new function is either constant, in which case the theorem is obvious, or it goes above 0 at some point (If not prove the theorem for minus the function). Since it is continuous on a closed, bounded interval, it is bounded, and has either a maximum or a minimum. At such a point, the derivative exists by our hypothesis, and it can't possibly be something other than 0. Why? Suppose the derivative were positive at a maximum point c, then by definition of limits, $\frac{f(x)-f(c)}{x-c}$ would be positive if we make x close enough to c with $x>c$, implying $f(x)>f(c)$ which is a contradiction.

In particular, we have Rolle's theorem, which says that if $f(a) = f(b) = 0$ and f is continuous on $[a, b]$ and differentiable on (a, b) then there is a point $a<x<b$ such that $f'(x) = 0$

□

Lemma. Let f be continuous on $[a, b]$ and n-1 times continuously differentiable, but also n times differentiable on the open interval (a,b), and suppose also that

$$f(a) = f'(a) = f''(a) = \cdots = f^{(n-1)}(a) = f(b) = 0$$

then there is some x in (a,b) such that $f^{(n)}(x) = 0$.

Proof. Induct on n. The $n = 1$ base case is just Rolle's theorem. Suppose we have $k < n$ and $x_k \in (a, b)$ such that $f^{(k)}(x_k) = 0$. Since $f^{(k)}(a) = 0$, we can find $x_{k+1} \in (a, x_k)$ such that $f^{(k+1)}(x_{k+1}) = 0$ by Rolle's theorem. So the result follows by induction.

□

Example. The function $x^3 - x^2$ has its first derivative and value equal to 0 when $x = 0$, and its value equal to 0 at $x = 1$, so between 0 and 1 it must have a point of second derivative zero. Indeed it does when $x = \frac{1}{3}$.

[2]Perhaps this is surprising, since if you try to make a discontinuous derivative, then unless you are really smart about it you will introduce a cusp that makes the function not differentiable. In Analysis we will see how this can be done. Hint: Consider the derivative of a function that is 0 at 0 and $x^2 \sin\left(\frac{1}{x}\right)$ elsewhere and consider what happens near 0.

Proposition. If $f(x)$ is n times differentiable at x_0 then

$$R(x) := f(x) - f(x_0) - (x - x_0) f'(x_0) - \frac{(x-x_0)^2}{2!} f''(x_0) - \cdots - \frac{(x-x_0)^n}{n!} f^{(n)}(x_0)$$

is $o((x-x_0)^n)$ as $x \to x_0$.

Proof. The first n derivatives of the above expression are all 0 when evaluated at x_0 because for k between 0 and n inclusive the $\frac{(x-x_0)^k}{k!} f^{(k)}(x_0)$ term has k'th derivative equal to $f^{(k)}(x_0)$ and all its other derivatives are 0 at x_0 by repeated application of the power rule so therefore it cancels the k'th derivative of the $f(x)$ term to make it equal to 0.

So, by the definition of o, we need to show that $\frac{R(x)}{(x-x_0)^n} \to 0$ as $x \to x_0$. This is fine as $(x-x_0)^n$ is not 0 when you make it arbitrarily small so it can go in the denominator.

Now, if we can show that any function whose first n derivatives are all 0 at x_0 is $o((x-x_0)^n)$, we will be done. It is clear that a function whose value and first derivative is 0 at x_0 is $o(x-x_0)$ since $\frac{g(x)}{x-x_0} = \frac{g(x)-g(x_0)}{x-x_0} \to g'(x_0) = 0$, so this theorem is true if n=1. But now suppose that we know that this theorem is true for n=k, then we will prove it is also true for n=k+1, therefore proving this by induction. Let g(x) be such that at x_0 its value and first k+1 derivatives are all 0. By the induction hypothesis, g'(x) has its first k derivatives and value equal to 0 at x_0 so $g'(x_0+h) = h^k \varepsilon(h)$ with $\varepsilon(h) \to 0$ as $h \to 0$, by the definition of little o notation and the induction hypothesis.

By the mean value theorem for some θ between 0 and 1 we have $hg'(x_0+\theta h) = g(x_0+h) - g(x_0)$ and $g'(x_0+\theta h) = (\theta h)^k \varepsilon(h)$, meaning that by the induction hypothesis

$$g(x_0+h) - g(x_0) = h\left((\theta h)^k \varepsilon(\theta h)\right) = \theta^k h^{k+1} \varepsilon(\theta h) = o(h^{k+1})$$

So induction done and proof complete.

□

Remark. Just because each remainder is small does not mean that the taylor series converges, since convergence at some x requires that there, the remainder is uniformly small, but it could be that the interval where the remainder is small shrinks and is zero in the limit. There are functions such that the taylor series does not converge in any interval despite the functions being infinitely differentiable (these are difficult to construct explcitly but the point is it is not automatic that a taylor series that is convergent exists), or functions where the taylor series converges but does not equal the function, like $\exp(-x^{-2})$ around x=0. However, it turns out that if the function has a complex derivative the taylor series always converges in some interval, see complex analysis in a later level. I'm not proving it because we don't need to use it for this purpose, but it trivializes all of these results.

Lemma. Let f be n+1 times differentiable on [a,b] with it's (n+1)'th derivative continuous. Then we have that

$$f(b) = f(a) + (b-a) f'(a) + \frac{(b-a)^2}{2!} f^{(2)}(a) + \cdots + \frac{(b-a)^n}{n!} f^{(n)}(a) + \int_a^b \frac{(b-t)^n}{n!} f^{(n+1)}(t)\, dt$$

Proof. We will do this by induction on n. When n=0 the theorem says

$$f(b) = f(a) + \int_a^b f'(t)\, dt$$

Which is known to be true. Now we will do integration by parts to get

$$\int_a^b \frac{(b-t)^n}{n!} f^{(n+1)}(t) dt = \left[\frac{-(b-t)^{n+1}}{(n+1)!} f^{(n+1)}(t)\right]_a^b + \int_a^b \frac{(b-t)^{n+1}}{(n+1)!} f^{(n+2)}(t) dt$$

$$= \frac{(b-a)^{n+1}}{(n+1)!} f^{(n+1)}(a) + \int_a^b \frac{(b-t)^{n+1}}{(n+1)!} f^{(n+2)}(t) dt$$

So the lemma follows by induction.

□

Proposition. If $f(x)$ is n+1 times continuously differentiable in an open interval containing x_0 then

$$f(x)-f(x_0)-(x-x_0)f'(x_0)-\frac{(x-x_0)^2}{2!}f''(x_0)-\cdots-\frac{(x-x_0)^n}{n!}f^{(n)}(x_0)$$

is $O((x-x_0)^{n+1})$ as $x\to x_0$.

Proof. Let $x_0=a$ and let b be such that the derivative is continuous on $[a,b]$ and let $x\to a$. The lemma above immediately implies the result since $f^{(n+1)}$ is continuous on the closed interval $[x_0,x]$ and thus bounded there, say it is bounded in absolute value by M. Then the integral is at most $\int_a^x \frac{(x-t)^n}{n!}M dt = M\frac{(x-a)^{n+1}}{n+1!}$ which is $O((x-a)^{n+1})$ as required. As we

□

Proposition. $f(x)-f(x_0)-(x-x_0)f'(x_0)-\frac{(x-x_0)^2}{2!}f''(x_0)-\cdots-\frac{(x-x_0)^n}{n!}f^{(n)}(x_0)=\frac{(x-x_0)^{n+1}}{(n+1)!}f^{(n+1)}(t)$ for some t(x) between x_0 and x, provided the first n+1 derivatives of f exist and are continuous from x_0 to x. (In fact we don't need the $n+1$'th derivative to be continuous, but merely for it to exist on the open interval (x_0,x) as that is enough to apply Rolle's theorem).

Proof. The n=0 case is just the mean value theorem.

Consider $g(x)\coloneqq f(x)-f(x_0)-(x-x_0)f'(x_0)-\frac{(x-x_0)^2}{2!}f''(x_0)-\cdots-\frac{(x-x_0)^n}{n!}f^{(n)}(x_0)$. This has its value and first n derivatives at x_0 equal to 0 by the same reasoning as we gave in one of the earlier proofs

Set $\frac{g(x)}{(x-x_0)^{n+1}}=C$ when x is our constant. Then $g(x)-C(x-x_0)^{n+1}$ as a function of x (so x is no longer constant) is 0 at x from how we defined C and has its first n derivatives and value equal to 0 at x_0 meaning it satisfies the conditions for generalized rolle's theorem. Therefore, there exists a point y between x_0 and x such that the n+1'th derivative of $g(y)-C(y-x_0)^{n+1}$ is 0. But the derivative of the second term is just C(n+1)! by the power rule, thus the n+1'th derivative of g at y is also C(n+1)!, thus $C=\frac{g^{(n+1)}(y)}{(n+1)!}$. Therefore, since we have that $g(x)-C(x-x_0)^{n+1}$ is 0 at x from earlier, we thus have that

$$g(x)=C(x-x_0)^{n+1}=\frac{g^{(n+1)}(y)}{(n+1)!}(x-x_0)^{n+1}$$

Which is exactly what we wanted to prove.

□

Example. There exists some point between 0 and 1 such that $\frac{e^y}{2}=e-2$, because if $x_1=1$ and x_0 =0, then $e^{x_1}-x_1-1=\frac{x_1^2}{2}e^y$ for some y between 0 and 1, but x=1 so this simplifies to $\frac{e^y}{2}=e-2$.

6.3 L'hopital's rule

Lemma. (Cauchy mean value theorem) Let f,g be functions taking real values continuous on $[a,b]$ and differentiable on (a,b). Then there exists a c in (a,b) such that $g'(c)(f(b)-f(a))=f'(c)(g(b)-g(a))$

Proof. Try the function $h(x)=[g(x)-g(a)][f(b)-f(a)]-[g(b)-g(a)][f(x)-f(a)]$.

Now $h'(x)=g'(x)(f(b)-f(a))-f'(x)(g(b)-g(a))$

This is 0 at some point because $h(a)=h(b)$ and Rolle's theorem is true.

□

Proof of L'hopital's rule

Proof. Suppose that the following hold:

1. $f(a) = g(a) = 0$

2. $g(x)$ and $g'(x)$ is not 0 if x is close to a

3. $\lim_{x\to a^+} \frac{f'(x)}{g'(x)}$ exists and is equal to l.

4. f and g are differentiable in some open interval $(a, a+\eta)$ and continuous at a.

By the Cauchy mean value theorem $g'(t)(f(b)) = f'(t)(g(b))$ for some t between a and b, for any $a + \eta > b > a$. Rearranging gives that

$$\frac{f(b)}{g(b)} = \frac{f'(t)}{g'(t)}$$

I emphasise that t depends on b but for each fixed b it exists and is between a and b. Therefore, as $b \to a$, so does t.

Thus, $\frac{f(b)}{g(b)} \to l$ as $b \to a$.

If instead the limit is from the left the proof is analogous.

□

7 Uniform convergence

We will define uniform convergence.

Here are the definitions.

Convergence: At every x, for every $\varepsilon > 0$ there exists an $N > 0$ such that $n > N$ implies $|f_n(x) - f(x)| < \varepsilon$

Uniform Convergence: F or every $\varepsilon > 0$ there exists an $N > 0$ such that for every x, $n > N$ implies $|f_n(x) - f(x)| < \varepsilon$

We know about how pointwise convergence works – a sequence of functions converges to a function if it converges at every value. We can say "For each point, there is an $\varepsilon > 0$ such that for all x, our sequence of functions $f(x)$ is eventually, after some n, within ε of the limit function". This is different from uniform convergence in the sense that uniform convergence requires n not to depend on x. We need the function to eventually get arbitrarily close at all points at once, not just at each point. An example of a sequence of functions that converges pointwise but fails this is the following:

$$f_n(x) = \begin{cases} 0 < x < \frac{1}{n} : 1 - nx \\ \text{Otherwise} : 0 \end{cases}$$

This looks like the functions in these images:

Figure 4 shows examples of this for n=1, 2, 10.

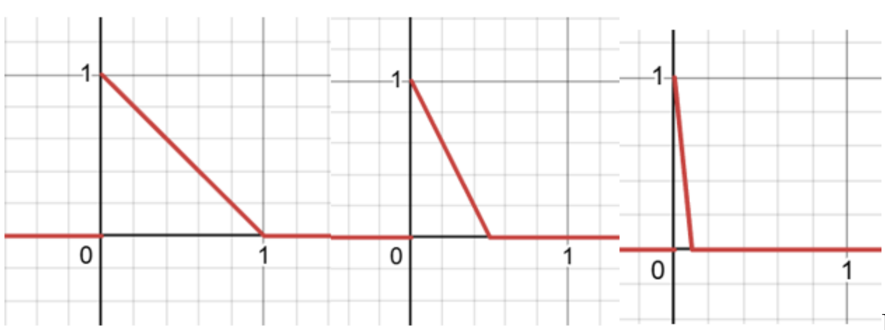


Figure 4

The point is: For any point you choose, say $k > 0$, once $n \geq \frac{1}{k}$, it will get to 0. So the functions converge pointwise to 0. It seems like we can pick $x = 0$ and it will converge to 1 but we defined $f(0)$ to be 0. We do not converge uniformly to 0 as we never satisfy that all points get arbitrarily close to 0.

It is useful because, for example, if we want to swap a limit with an integral we know that the total error will eventually get small, rather than just the error at each point.

The right mental picture of uniform convergence is that the function should be forced to be within a band around the limit eventually.